Networks, Categories, and Collaboration: The Emergence and Consolidation of Chilean *Música Urbana*

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Abstract

This dissertation examines how Chilean Música Urbana became relationally organised through Spotify-visible artist collaborations between 2014 and 2024. It compares two periods, formation and national consolidation (2014-February 2022) and boom and international expansion (March 2022-2024), using descriptive social network measures, weighted collaboration patterns, community detection, and binary exponential random graph models. The results show that the collaboration network expanded rapidly while becoming more structurally integrated, that collaboration remained unevenly distributed, and that communities were differentiated by generation, musical trajectory, country composition, and organisational position. The dissertation argues that Chilean Música Urbana should be understood not only as a genre label, but as a relationally organised music world shaped by platform visibility, generational differentiation, and selective internationalisation.

Declaration

This dissertation is my original work, except where otherwise clearly acknowledged and referenced.

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1. Introduction

Over the last decade, *Música Urbana* has become one of the most visible and commercially successful sectors of Latin American popular music (Muñoz-Tapia et al., 2025). Artists associated with the category have accumulated billions of streams, occupied prominent positions in digital rankings, and expanded their audiences across national borders. In Chile, this process accelerated during the early 2020s, when artists such as Cris Mj, Polimá Westcoast, Floyymenor, Pailita, Jere Klein, Paloma Mami and others achieved unprecedented levels of visibility both nationally and internationally. As a result, *Música Urbana* has become a central reference point for artists, audiences, media outlets, streaming platforms, and music industry actors.

The term *Música Urbana*, or *Género Urbano*, is retained in Spanish rather than translated as urban music in order to preserve its situated meaning within Latin American musical contexts. In Chile and across the region, *Música Urbana* functions as an emic category that encompasses genres such as reggaetón, trap, dembow, mambo chileno, and rap, while also referring to broader cultural practices, identities, and forms of participation (Muñoz-Tapia et al., 2025; Muñoz-Tapia and Pinochet-Cobos, 2025). Although the category is now widely used, its meaning remains contested. Artists grouped under the label often differ in musical style, trajectories, audiences, and aesthetic references, raising questions about what connects them and how such a heterogeneous category acquires social coherence.

Existing research has largely approached this question through the study of genres, cultural classifications, media discourse, platformisation, and the political economy of digital music (Muñoz-Tapia et al., 2025; Muñoz-Tapia and Pinochet-Cobos, 2025). These studies have shown that *Música Urbana* is neither a stable genre nor a simple continuation of earlier musical traditions. Rather, it emerged through processes of circulation, translation, categorisation, and institutional recognition that connected previously distinct musical practices. Building on Muñoz-Tapia and Pinochet-Cobos (2025), this dissertation takes the generational and categorical formation of Chilean *Música Urbana* as a starting point and asks how that category became relationally organised across time. The central contribution is therefore not to identify the generational divide itself, already described in previous research,

but to examine how this categorical and generational transformation became organized through collaboration networks.

This question is especially relevant in the context of streaming. As Hesmondhalgh (2025) argues, music streaming has produced ambivalent transformations. It has increased the international visibility of genres, scenes, and markets beyond the historical centres of the global music industry, particularly the United States and the United Kingdom, contributing to a more multipolar circulation of music. At the same time, this diversification has not eliminated structural inequalities. Platform ownership, digital infrastructures, algorithmic intermediation, and international intellectual property regimes continue to concentrate power within the global music economy. Chilean *Música Urbana* is a productive case for examining this tension: it has contributed to the formation of a relatively distinctive artistic world and local market, while remaining deeply articulated with transnational genres, international collaborations, platform-based visibility, and the commercial logics of streaming.

This dissertation argues that collaboration networks provide a useful way of examining these processes. Collaborations are not merely technical arrangements for producing songs. They are also mechanisms through which artists gain visibility, access new audiences, establish reputations, and create connections across musical communities. In this sense, collaboration networks make it possible to observe how heterogeneous genres and artists become articulated into broader musical formations. Rather than focusing exclusively on discourse or classification, this study examines the relationships through which *Música Urbana* is enacted in practice.

To do so, the dissertation analyses collaboration networks among artists associated with Chilean *Música Urbana* between 2014 and 2024 using data derived from Spotify metadata. The analysis compares two historical periods: a phase of formation and national consolidation (2014–February 2022) and a phase of boom and international expansion (March 2022–2024). Drawing on relational sociology, social network analysis, and research on music worlds, the study investigates how collaboration patterns changed between these periods as the scene expanded, became more structurally integrated, and internationalised selectively rather than uniformly.

The dissertation addresses three research questions. First, how did the structure of collaboration networks change between the formation and national consolidation period and the boom and international expansion period? Second, to what extent did processes of internationalisation transform both the composition and relational structure of collaboration networks? Third, how were collaboration opportunities distributed among artists as the scene expanded in scale, visibility, and international reach?

By addressing these questions, the dissertation contributes to research on music worlds, cultural classification, and digital cultural production. More broadly, it shows how social network analysis can be used to investigate the relational transformation of musical categories over time. In this case, the comparison between periods makes it possible to move from the question of categorical emergence to the question of how *Música Urbana* expanded, integrated, and internationalised as a collaboration network.

2. Theoretical Framework

2.1 Categories and Relations: Musical Assemblages, Social Identity, and Social Network Analysis

Musical genre can be understood as a historically situated category that classifies musical practices while linking them to particular audiences, identities, institutions, and modes of circulation. Following Brackett (2016), genres do not possess fixed essences; they acquire meaning relationally, through their position within changing systems of musical difference. Brackett develops this argument through a reading of genealogical, citational, and dialogical approaches associated with Foucault (1977), Derrida (1988), and Bakhtin (1981), while also building on Fabbri's theory of musical genres as systems of conventions and diachronic processes (Fabbri, 1982, 2012). From this perspective, genres persist because works repeat, cite, and transform recognisable conventions over time. Genre is therefore not simply a label attached to sound, but a historically produced way of organising recognition.

This approach links genre to a conception of assemblage. Brackett (2016) and Drott (2013) both understand genres as configurations of heterogeneous elements connected through

relations of exteriority: sounds, images, performance styles, modes of dress, forms of speech, technologies, institutions, markets, media, and classificatory devices. A genre is therefore not reducible to musical style alone. It is an arrangement of sonic, social, institutional, and technological elements that can be reconfigured over time. This point is important for the present dissertation because networks of artist collaboration on songs can be considered part of a broader set of relations through which musical categories are produced and recognised.

Musical categories also organise forms of identification. Brackett (2016), drawing on Born (2000, 2011), shows that genres become associated with social groups through historically contingent processes involving class, race, ethnicity, gender, generation, and market position. These associations are neither natural nor stable. They are produced through circulation, reception, media discourse, institutional classification, and artistic practice. Genre categories therefore help constitute musical publics: they provide names, expectations, and boundaries through which actors understand themselves and others. In this sense, musical categories are not merely descriptive; they are performative classifications that shape how cultural participation becomes socially intelligible.

From another perspective, Social Network Analysis and relational sociology have also addressed the relationship between categories and relations. White's concept of catnets is central here. For White (2008), social formations emerge through the intersection of categorical identities and network ties. Categories allow actors to recognise one another as belonging to a shared formation, while relations provide the channels through which those categories circulate, stabilise, and become consequential. This framework is especially useful for studying musical worlds because artists may share a category without all being directly connected, and they may become connected in ways that subsequently reshape the meaning of the category itself.

Building on White and Tilly, Crossley (2025a) develops this point by linking Social Network Analysis with Social Identity Theory. Social Identity Theory (Tajfel and Turner, 1979) explains how actors construct shared identities through categories, norms, and prototypes, and how these classifications orient perception, recognition, and affiliation. However, it tends to under-theorize the relational channels through which identities circulate and acquire durability. Social Network Analysis, by contrast, explains how homophily, closure,

brokerage, diffusion, and network position structure interaction, but often gives less attention to the symbolic categories through which actors interpret those relations. Crossley's contribution is to bring these two perspectives together.

From this perspective, categories and networks are mutually constitutive. Categories circulate through networks and become socially effective when they are recognized, shared, and enacted by connected actors. At the same time, categories shape networks by guiding affiliation, recognition, in-group preference, and expectations about appropriate ties. Actors do not collaborate only because they occupy positions in a structure; they also interpret one another through categories that make some collaborations more thinkable, legitimate, or desirable than others. Conversely, repeated relations can reinforce, modify, or destabilize the categories through which actors understand the field. This retroactive relation between categories and ties is central to the dissertation's argument.

Social Network Analysis therefore provides a methodological strategy for examining one dimension of this process. It does not explain musical categories on its own, nor does it reduce genres to networks. Its contribution lies in making relational configurations visible: who collaborates with whom, how ties are distributed, which actors occupy central or peripheral positions, how clusters form, and whether collaboration follows patterns of generational proximity, national boundary-making, or categorical similarity. These patterns require historical and cultural interpretation, but they provide empirical evidence about how categories become organized through relations.

This is important considering the ambiguity and fluidity of a category such as *música urbana*. In that sense, the notion of hypergenre is useful for understanding categories that gather heterogeneous musical forms without eliminating internal differences. Buch's work on hypergenre describes broad musical categories that organise diverse practices under a shared classificatory umbrella (Buch, 2013, 2018). *Música Urbana* can be approached in these terms. Rather than naming a bounded genre, it operates as a glocal hypergenre that connects rap, trap, reggaeton, dembow, mambo chileno, RKT, and other local or transnational variants (Muñoz-Tapia et al., 2025). Its coherence does not depend on a single musical essence, but on the capacity of the category to connect heterogeneous sounds, actors, scenes, markets, and infrastructures.

Previous historical and ethnographic research on Chilean *Música Urbana* has shown that the category emerged through interactions among artists, media, platforms, audiences, industry actors, local scenes, and global tendencies (Muñoz-Tapia et al., 2025; Muñoz-Tapia and Pinochet-Cobos, 2025). This earlier work adopted a perspective close to assemblage approaches by reconstructing the historical formation of *Música Urbana* as a category and identifying the meanings, tensions, and divisions that structure the Chilean case. In particular, it highlighted the importance of generational differentiation between historical rap trajectories and younger *urbano* formations. The present dissertation builds on that interpretation but shifts the empirical focus toward artist collaboration on songs, operationalised as Spotify-visible co-presence among credited performers on the same track. It examines how those historically identified divisions are organised relationally, while also incorporating additional attributes such as internationalisation, gender, artist type, and record-label position.

In this dissertation, *Música Urbana* is therefore analysed simultaneously as a category and as a relational formation. The category provides a historically situated framework of recognition, while networks of artist collaboration on songs reveal how performers associated with and around that category became connected across time. This approach allows the dissertation to examine how Chilean *Música Urbana* was not only named, narrated, or classified, but also enacted through credited collaboration. The central question is not whether *Música Urbana* is a "real" genre in a bounded stylistic sense, but how it became socially effective as a category through patterned relations that sustained, expanded, and differentiated it.

2.2 Collaboration Networks and Music Worlds

Scholars working on music, cultural production, and collaboration have consistently emphasised that music is a collective activity rather than the product of isolated creators. Small's (1998) concept of *musicking* treats music as a relational activity involving performers, listeners, mediators, and facilitators, while Becker's (2008) theory of art worlds understands artistic production as a cooperative system coordinated through conventions among creators, technicians, distributors, critics, and audiences. Farrell's (2001) notion of

collaborative circles similarly highlights how sustained peer interaction can generate innovation and mutual influence.

Crossley develops this argument through the concept of music worlds (Crossley, 2009, 2015, 2020, 2025b; Crossley et al., 2015). Music worlds are relational formations in which musical practices are produced, circulated, and sustained. His contribution is both theoretical and methodological: collaborations, shared spaces, reputations, and recurrent interactions generate the relational infrastructure through which scenes emerge and endure. Social network analysis therefore complements sociohistorical interpretation by allowing researchers to identify the structures through which music worlds acquire coherence.

Research on music networks shows that collaboration is not merely a technical arrangement for producing songs. It is also a mechanism through which music worlds are formed, scenes acquire cohesion, and categories become socially credible. Kirschbaum and Carvalho de Vasconcelos (2007) analyse Tropicalia as the outcome of strategic connections across previously separated musical communities, while Hammou (2009) shows that French rap developed as a professional social world through a cohesive network of featuring collaborations. Similarly, Grubesic et al. (2024) demonstrate that Yacht Rock depended upon an interconnected network of musicians embedded within the social and spatial infrastructure of Los Angeles.

Collaboration networks also influence artistic careers and forms of visibility. Network position may affect artistic success (Silva et al., 2019), increase future streaming visibility (Deshmane and Martinez-de-Albeniz, 2020), and generate reputational associations among artists and composers (Braden et al., 2024). Network evolution also matters: Bagley et al. (2022) identify processes of fragmentation and recombination in musical collaboration networks, while Xu et al. (2025) show that social ties can expand the range of musical preferences available to individuals.

A further contribution of this literature concerns the relationship between local and external ties. Makkonen (2017) argues that local networks generate opportunities and support career development, whereas external connections are often more important for broader visibility and international projection. Similarly, Diani and Sacchetti (2023) show that artistic vitality tends to increase when musicians combine local embeddedness with external connections.

This distinction is central for the present study because the question is not only whether Chilean *Música Urbana* expanded, but also whether its collaboration structure became more internationally connected.

Digitalisation has transformed these relational dynamics. According to Makelberge (2012), technological changes in production infrastructures modify the conditions of collaboration by enabling creative interaction beyond shared physical spaces. In contemporary music worlds, collaborations increasingly occur through mediated environments in which digital platforms, metadata, and visibility metrics become part of the relational process itself.

The featuring is therefore a particularly useful relation for studying *Música Urbana*. Unlike friendship, influence, or informal cooperation, featured collaborations are public, credited, platform-visible, and cumulative. They create observable traces of association between artists that can be systematically reconstructed through digital metadata. While featuring collaborations do not capture the full social life of musical production, they provide a meaningful relational layer through which patterns of cooperation, recognition, and visibility become empirically observable.

3. Research Background and Hypothesis Formulation

3.1 From Urban Contemporary to Chilean *Música Urbana*

The emergence of *Música Urbana* in Chile forms part of broader transformations in the categorisation of popular music. One important antecedent is Urban Contemporary, which emerged in the United States during the 1970s and 1980s as a radio and industry classification grouping diverse forms of Black popular music (Hammou, 2023; Muñoz-Tapia et al., 2025). A related process occurred within Latin music industries, where labels such as urban, Latin urban, reggaeton, and Latin rap became associated through shared markets, media circuits, and cultural imaginaries (Rivera-Rideau, 2015). These categories did not describe unified musical styles. Rather, they assembled heterogeneous genres through infrastructures of circulation, audiences, and institutional classification.

This dissertation approaches *Música Urbana* in Chile as a situated category related to, but not identical with, those classifications. Categories such as Urban Contemporary, hurban, Latin Urban, and *Música Urbana* function as *false friends*: they appear equivalent because they share labels, references, and classificatory functions, while in practice organising different musical worlds and historical trajectories (Muñoz-Tapia et al., 2025). At the same time, they reveal elective affinities among heterogeneous genres such as rap, reggaeton, trap, dembow, mambo chileno, and RKT. These affinities are not only musical but also social, since urban categories have frequently grouped cultural expressions associated with racialised populations, working-class sectors, migrants, Indigenous peoples, and ethnic minorities. They do not eliminate difference; rather, they create conditions under which diverse musical practices can be recognised as part of a broader cultural formation.

The Chilean case reflects these broader dynamics while following its own historical trajectory. Since the late 1980s and early 1990s, Hip Hop scenes established local infrastructures of production, circulation, and artistic socialisation, contributing to independent forms of cultural production that later shaped the emergence of urban genres (Muñoz-Tapia and Pinochet-Cobos, 2025). Contemporary *Música Urbana*, however, cannot be reduced to a continuation of Chilean rap. During the 2010s, digital production technologies, social media, streaming platforms, and digital distribution services facilitated the growth of trap, reggaeton, mambo chileno, and related genres among younger generations. By the early 2020s, *Música Urbana* had become a widely recognised category capable of grouping these heterogeneous trajectories under a common label (Muñoz-Tapia et al., 2025; Muñoz-Tapia and Pinochet-Cobos, 2025).

At the same time, *Música Urbana* maintained a complex relationship with rap. In Chile and much of Latin America, rap has traditionally been associated with Hip Hop culture, boom bap aesthetics, and underground scenes (Muñoz-Tapia and Pinochet-Cobos, 2025). Unlike the United States, where rap or Hip Hop often function as broader categories that encompass styles such as trap, *Música Urbana* in Latin America and Spain emerged as a wider umbrella that groups rap alongside genres with distinct trajectories, including reggaeton, trap, and dembow. Within this configuration, reggaeton has played a particularly important role in

articulating connections between otherwise heterogeneous genres and audiences within the broader metageneric structure of *Música Urbana* (Arias Salvado, 2025).

Rather than examining this process through discourse alone, this dissertation analyses the collaboration networks through which these connections were enacted. If *Música Urbana* emerged through the articulation of heterogeneous musical traditions, collaboration networks provide a way to examine how those articulations became structured over time.

3.2 Spotify, Collaboration and Platform Visibility

Spotify occupies a central position in this study because it functions simultaneously as a platform for music circulation and as a source of relational data. This is particularly relevant for Chilean *Música Urbana*, whose expansion has been closely associated with streaming platforms, social media, digital distributors, and platform-mediated forms of visibility.

This role should be situated within broader processes of digital transformation. A first phase was characterised by disintermediation, as home studios, MP3 technologies, websites, blogs, peer-to-peer systems, and social media reduced barriers to recording, distribution, and promotion, allowing artists to circulate music outside traditional industry structures (Albornoz, 2011; Buquet, 2003; Fouce, 2012; Gallo and Semán, 2016; Leyshon, 2001). In Chile, these developments coincided with the consolidation of rap as a largely underground and independent musical world (Muñoz-Tapia and Pinochet-Cobos, 2023, 2025).

A second phase emerged with the rise of digital streaming platforms. Rather than eliminating intermediaries, digitalisation created new forms of reintermediation, in which distributors, playlists, recommendation systems, metrics, algorithms, and major labels became central to organising visibility and audience reach (Lamacchia, 2016; Leyshon, 2001; Negus, 2019; Prey, 2016). Spotify is therefore not simply a distribution channel but part of a broader ecosystem through which artistic visibility and commercial value are organised.

For the purposes of this dissertation, Spotify is important not only as an industry actor but also as an empirical source. Track-level collaborations constitute public, credited, and platform-visible relations between artists. Although they do not capture the full range of creative and organisational relationships involved in music production, they provide

observable traces of artistic association that can be systematically reconstructed through metadata.

3.3 Periodisation and Hypotheses

This dissertation examines collaboration networks during a period of rapid transformation in Chilean *Música Urbana*. To capture these changes, the analysis distinguishes between two historical phases. The first, Formation and National Consolidation (January 2014–February 2022), corresponds to the emergence and consolidation of *Música Urbana* as a recognisable category within Chile. During this period, artists associated with rap, trap, reggaeton, and related genres increasingly interacted through collaborations, digital platforms, and shared audiences.

The second phase, Boom and International Expansion (March 2022–December 2024), corresponds to a period of accelerated growth and transnational visibility. This period was marked by the international success of Chilean artists, increased presence in global streaming rankings, growing industry attention, and more collaborations with foreign artists. Songs such as *Una Noche en Medellín*[1] and *Ultra Solo*[2] became emblematic of this broader transformation.

The distinction between these periods should be understood as an analytical device rather than a rigid historical rupture. Processes of internationalisation were already present before 2022, while domestic collaborations remained important afterwards. Nevertheless, the periodisation captures an important shift in the scale, visibility, and geographical reach of the scene.

Four hypotheses guide the empirical analysis.

H1. Network Expansion

[1] See: https://www.youtube.com/watch?v=EaIh7tRqdqM&list=RDEaIh7tRqdqM&start_radio=1

[2] See: https://www.youtube.com/watch?v=Mub403hj9nE&list=RDMub403hj9nE&start_radio=1

The boom and international expansion period will contain larger collaboration networks, characterised by a greater number of artists and collaborations and by stronger component-level integration.

H2. Internationalization

The boom and international expansion period will contain a greater proportion of foreign artists and a larger number of cross-national collaborations, although these ties may remain selectively concentrated rather than generalised across the whole network.

H3. Tie Intensification

The boom and international expansion period will contain a higher proportion of recurrent collaborations, reflecting the consolidation of collaborative relationships within the scene.

H4. Uneven Collaboration Participation

Despite overall expansion, collaborations will remain unevenly distributed, with a minority of artists accumulating substantially more ties while many artists enter the network through only one or a few collaboration partners.

These hypotheses treat changes in network structure as indicators of broader transformations in the organisation of Chilean *Música Urbana*. The following section describes the data sources, network construction procedures, and analytical strategies used to evaluate these expectations.

4. Data and Methods

4.1 Data Sources and Extraction

The collaboration networks were constructed from a larger database of Chilean musicians and their Spotify-visible collaborations. Data collection combined two complementary sources. The primary source was the Spotify Web API, accessed through the Spotipy Python library using Spotify's Client Credentials workflow. The extraction retrieved artist metadata, releases, tracks, credited performers, and release dates. Historical information on tracks

appearing in the Spotify Top 200 chart for Chile was retrieved from Kworb and used as complementary evidence of platform visibility and ranking presence.

The Spotify extraction began from manually curated playlists focused on Chilean music, which were screened to retain lists composed of Chilean artists and used as seed lists for identifying national performers. From this seed set, the Python pipeline retrieved artist identifiers and metadata, queried releases, and extracted track-level performer credits.

Collaborations were generated at the track level. Whenever a Spotify track listed more than one credited performer, all unordered artist pairs among those performers were created. Each collaboration record therefore contains two Spotify artist identifiers, a Spotify track identifier, and the release date. The resulting networks capture public, credited co-presence on Spotify releases rather than songwriting, production, management, friendship, or live-performance relationships. Because the data indicate joint participation on a track, the networks are undirected.

The extracted files should be understood as historical source data rather than as a continuously updated representation of Spotify. Platform metadata can change over time, and the original API responses were not archived as immutable snapshots. The repository preserves the extracted collaboration tables and processing logic used in this dissertation, but it cannot reconstruct later changes made by Spotify to artist profiles, genre labels, release credits, or follower counts.

4.2 Network Construction and Delimitation

The initial collaboration database spans multiple genres, scenes, and historical trajectories. Rather than beginning with a fixed list of artists, the network was delimited around a broad Chilean *Música Urbana* seed category while preserving relevant ties to adjacent musical fields and international collaborators. This approach reflects the assumption that *Música Urbana* is best studied as a relational, historically situated category rather than a fixed stylistic genre.

Network delimitation proceeded in three steps. First, Spotify's genre metadata provided an inclusive automated screening. Artists were initially coded as potential *Música Urbana* cases

when their genre field contained at least one of the following terms: reggaeton, hip hop, trap, rap, mambo, urbana, urbano, neoperreo, r&b, guaracha, dancehall, rkt, neo soul, acid jazz, or dembow. These labels were used only to construct the initial candidate universe, not as definitive genre categories.

Second, we reviewed the preliminary classification artist by artist. Spotify's platform labels were not treated as definitive because they inconsistently combine stylistic descriptions, national scenes, commercial categories, and algorithmically generated associations. Manual validation drew on previous research on *Música Urbana* and related musical genealogies in Chile, together with contextual knowledge of artist trajectories, repertoires, collaborations, and public positioning. This procedure allowed artists omitted or ambiguously classified by Spotify to be included while removing false positives generated by the broad keyword search.

Third, the manually validated Chilean *Música Urbana* seed was expanded one step outward. To enter the analytic network, Chilean seed artists had to participate in at least one retained Spotify collaboration released between 1 January 2014 and 31 December 2024. Tracks released outside this observation window were not counted. Foreign artists and non-urban artists were retained whenever they collaborated directly with the Chilean *Música Urbana* seed during the same period.

Nodes represent credited performers appearing on Spotify releases and include both solo artists and artistic groups or collectives. The final analytic network excludes entities that primarily operate as record labels, imprints, distributors, publishing companies, studios, booking agencies, or service brands. It also excludes duplicate collective accounts when including them would reproduce collaborations already represented by individual members. Before constructing the final networks, the manually validated artist database was restricted to profiles with more than 1,000 Spotify followers. These exclusions delimited the network; the subsequent analysis examines how artists classified within and beyond this boundary are connected through Spotify-visible collaborations.

4.3 Node Attribute Generation

Node attributes were generated through a two-stage process. First, the Spotify Web API provided artist identifiers and platform metadata, including follower counts, popularity

scores, and Spotify genre labels. Because Spotify does not provide standardised historical or biographical information, the database was subsequently enriched through an AI-assisted information retrieval workflow using Google's Gemini Flash 2.0 language model with web search. This workflow retrieved publicly available information on artist trajectories, places of origin and artistic formation, debut releases, career milestones, record-label affiliations, gender, and group membership composition.

The AI-assisted stage was used for retrieval and organisation rather than for final classification. Each extracted attribute was accompanied by a confidence rating and supporting sources when available. Final coding decisions were manually reviewed, corrected, and harmonised before inclusion in the analytical database.

4.4 Variable Operationalisation

The validated information was transformed into analytical variables suitable for network analysis. The principal node attributes used in the descriptive and modelling analyses are country, artist type, gender, record-label affiliation, and generation. The broad *Música Urbana* classification was used primarily to delimit the analytical network, while the network analyses examine the relational patterns produced by collaboration ties.

Generation was operationalised from the year of professional musical inception. For descriptive and community analyses, this year was grouped into cohorts: decade-based categories for artists beginning before 2000 and five-year intervals from 2000 onward. For ERGM models, the same inception year was used as a continuous attribute, centred within each period, in order to estimate the effect of temporal proximity between artists. When the inception year could not be established, we used the release year of the earliest Spotify track as a proxy. Record-label affiliation was classified into hierarchical categories distinguishing major labels, major subsidiaries, independent labels, and cases without an identifiable label. When artists were affiliated with more than one label type within the same observation period, the highest-ranking category was retained. For groups and collectives, gender was operationalised according to membership composition and subsequently harmonised into broader analytical categories where appropriate.

4.5 Weighted Networks and Periodisation

The networks are initially weighted. For each period, artist identifiers were converted into canonical unordered pairs, so that (A)-(B) and (B)-(A) represented the same dyad. Track-level records were then deduplicated using the combination of unordered dyad and Spotify track identifier. This prevents repeated extraction of the same track from inflating collaboration frequencies while preserving different tracks credited to the same pair. After canonicalization and deduplication, the final edge lists contain no self-loops, reverse-orientation duplicates, or repeated dyads.

The raw weight of a tie equals the number of unique retained Spotify tracks on which two artists were jointly credited during a given period. Because collaboration frequencies are strongly right-skewed, raw counts were also recoded into three ordered categories: 1 = sporadic for one collaboration, 2 = weak recurrent for two collaborations, and 3 = recurrent/strong for three or more collaborations. This ordinal measure is used in the descriptive analysis of collaboration intensity, the network visualisations, and the community detection analysis. It preserves meaningful differences between one-time and recurrent collaborations while reducing the influence of a small number of highly prolific dyads.

For the ERGM analysis, the networks were further dichotomised. The main ERGM specification models whether at least one collaboration exists between two artists within a period. This binary strategy was selected because the valued and ordinal ERGM specifications were less stable for the dissertation's comparative modelling purposes. The weighted and ordinal measures are therefore retained for descriptive and community analyses, while the ERGM focuses on the presence or absence of collaboration ties.

Separate networks were constructed for each historical period. Collaborations were assigned to periods according to track release date, allowing comparison between the Formation and National Consolidation period (2014-February 2022) and the Boom and International Expansion period (March 2022-2024). The two networks are non-cumulative: collaborations observed in the first period are not automatically carried into the second period. Tracks with incomplete calendar dates were excluded in order to apply a uniform date-quality rule.

4.6 Network Scope

Tables 1A and 1B summarise the main stages of node and dyad delimitation. A node may be removed because it is absent from the manually validated artist database, lacks valid country or artist-type information, falls outside the permitted working categories, was manually excluded as an organisational or redundant profile, or has no direct retained collaboration with the Chilean *Música Urbana* seed during the relevant period. Foreign *Música Urbana* and non-urban artists remain in the final networks when they collaborate directly with the Chilean seed.

Table 1A. Node delimitation after final filtering.

Period	Initial nodes	Matched validated profiles	Eligible profiles	Final nodes	Retained
Formation and national consolidation	840	474	438	213	25.4%
Boom and international expansion	1432	876	803	564	39.4%

Table 1B. Dyad delimitation after final filtering.

Period	Initial edges	Final edges	Edges removed	Retained
Formation and national consolidation	1425	410	1015	28.8%
Boom and international expansion	3592	1579	2013	44.0%

The scope tables are included to document the boundary-making procedure. Substantive comparisons of network size, connectivity, composition, and tie intensity are presented in the Results section.

4.7 Analytical Strategy

Although this dissertation relies primarily on quantitative network data, the analytical strategy is informed by previous historical and ethnographic research on the emergence of Chilean Música Urbana (Muñoz-Tapia and Pinochet-Cobos, 2025). That work helps identify the meanings, categories, and divisions sociologically relevant to interpreting the network results, especially the generational distinction between historical rap trajectories and younger urbano formations. In this sense, the study follows a mixed-methods logic rather than a fully independent mixed-methods design: qualitative knowledge derived from interviews,

participant observation, and historical reconstruction informs the construction and interpretation of quantitative network evidence (Domínguez and Hollstein, 2011).

All analyses were conducted in R using RStudio. Data cleaning, recoding, and table construction were carried out mainly with tidyverse packages, including dplyr, tidyr, readr, and stringr. Network objects and structural measures were produced using igraph, tidygraph, and network, while visualisations were generated with ggplot2, ggraph, and related plotting tools. Community detection was implemented through modularity-based algorithms, with particular attention to the Leiden solution, which was selected for its interpretable and stable meso-level clusters. ERGM models were estimated using the ergm and statnet framework, while goodness-of-fit and diagnostic procedures drew on ergm, btergm, and associated simulation tools. The dissertation was written and rendered with Quarto, allowing text, code, tables, and figures to be integrated in a reproducible workflow.

The analytical strategy proceeds in three stages. First, descriptive statistics compare the two period-specific networks in terms of size, density, degree distribution, connected components, centralisation, transitivity, and weighted collaboration intensity. These measures establish whether the network expanded, became more integrated, and changed in the distribution of collaboration activity. Second, community detection identifies meso-level collaboration clusters and examines how these clusters differ by generation, country composition, gender, artist type, label affiliation, and collaboration intensity. Third, binary ERGMs are used to evaluate whether selected structural and attribute-based mechanisms are associated with the probability of collaboration ties. The ERGM results are interpreted cautiously, not as causal explanations, but as formal tests of relational patterns suggested by the descriptive and community analyses.

5. Results

5.1 Descriptive Analyses[3]

The results first describe changes in the topology, composition, and tie intensity of the Spotify-visible collaboration network. These descriptive analyses establish the structural context for the subsequent community detection and ERGM analyses.

5.1.1 Network Topology

The first comparison asks how the network's global structure changed between periods. Tables 2A and 2B report binary network statistics, whereas Tables 3A and 3B summarise weighted measures of collaboration intensity, concentration, and closure.

Table 2A. Network size and connectivity by period.

Period	Nodes	Edges	Density	Components	Largest component	Largest component share
Formation	213	410	0.01816	9	191	89.7%
Boom	564	1579	0.00995	8	545	96.6%

Table 2B. Degree distribution and closure by period.

Period	Mean degree	Median degree	Maximum degree	Degree centralization	Transitivity
Formation	3.85	2	27	0.110	0.332
Boom	5.60	2	55	0.088	0.233

Table 3A. Weighted collaboration intensity by period.

Period	Total collaborations	Mean raw strength	Maximum raw strength	Mean ordinal strength	Maximum ordinal strength
Formation	605	5.68	66	4.74	40
Boom	2430	8.62	103	7.50	89

Table 3B. Weighted concentration and closure by period.

Period	Ordinal-strength centralization	Raw weighted transitivity	Ordinal weighted transitivity
Formation	0.070	0.326	0.321
Boom	0.039	0.405	0.401

[3] Code and supplementary materials are available at: https://github.com/sebastianmunozt/mu-ch

The Spotify-visible collaboration network expanded substantially. The number of artists increased from 213 to 564, while unique collaboration ties grew from 410 to 1,579. Total credited track collaborations increased from 605 to 2,430. Because the observation windows differ in length and platform visibility likely changed over time, these figures should be interpreted as descriptive network sizes rather than directly comparable rates or complete population estimates.

This growth was accompanied by greater structural integration. Mean degree increased from 3.85 to 5.60, while the median degree remained constant at two. The combination of a stable median with rising maximum degree and strength indicates increasing heterogeneity in collaboration activity. At the same time, degree centralisation declined from 0.110 to 0.088 and ordinal-strength centralisation declined from 0.070 to 0.039, suggesting that growth did not increase dependence on a single dominant actor.

The network also became less dense but more connected. Density declined from 0.0182 to 0.0100, while the largest connected component increased from 191 to 545 artists and from 89.7% to 96.6% of each network. Thus, a larger share of artists became connected within the same component even though a smaller fraction of all possible ties was realised.

Closure changed in different directions depending on whether ties are treated as binary or weighted. Binary transitivity decreased from 0.332 to 0.233, indicating fewer closed triangles relative to connected triples. In contrast, weighted transitivity increased using both raw counts and ordinal weights. The network therefore did not become more closed in the binary sense; rather, the closed local configurations that remained were characterised by greater collaboration intensity.

Figures 1 and 2 provide a visual summary of these changes. Node colour indicates country, node size is proportional to degree, and edge width and transparency indicate ordinal collaboration intensity. The first-period network appears organised around a smaller set of highly connected artists, whereas the second-period network displays a larger connected structure with more high-degree nodes and many peripheral artists attached through one or few ties. These layouts are descriptive rather than formal evidence of meso-level clustering, but they are consistent with the tabular results.

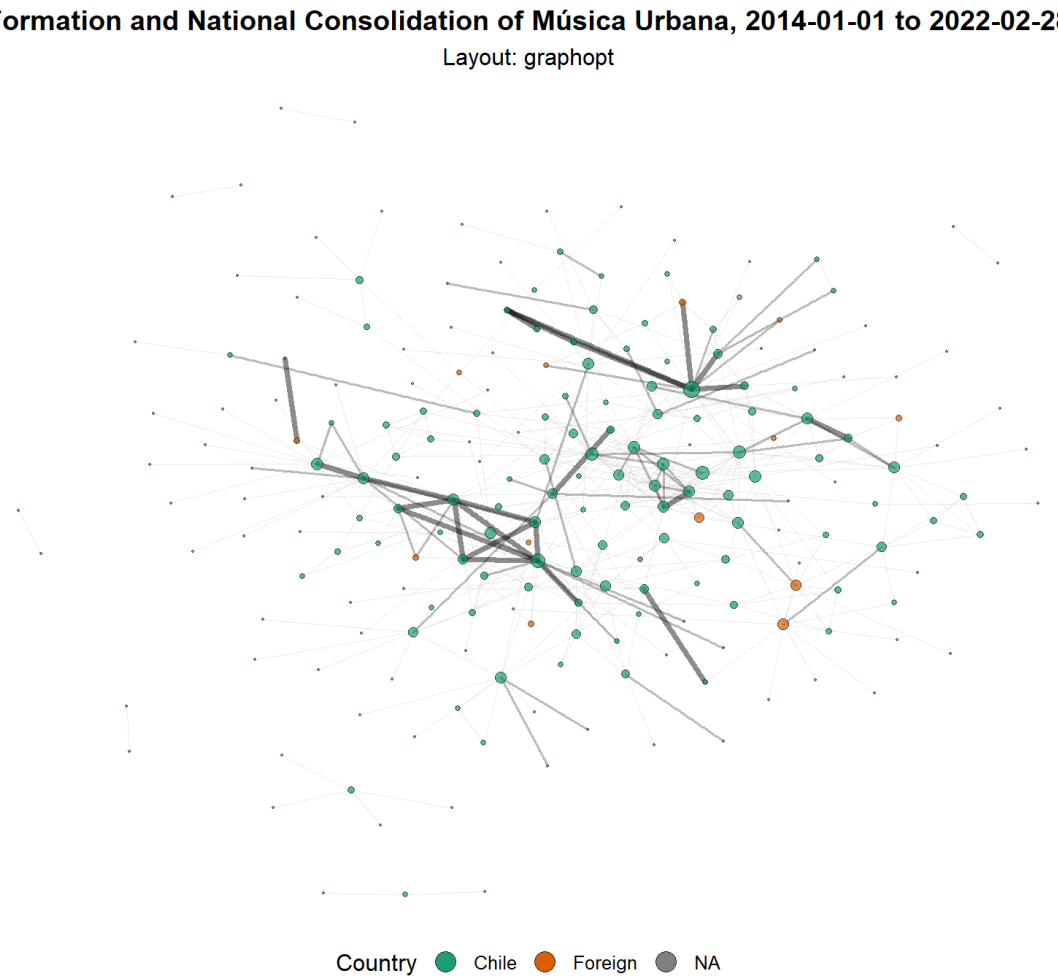


Figure 1. Collaboration network during the formation and national consolidation period. Node size is proportional to degree, defined as the number of distinct collaboration partners.

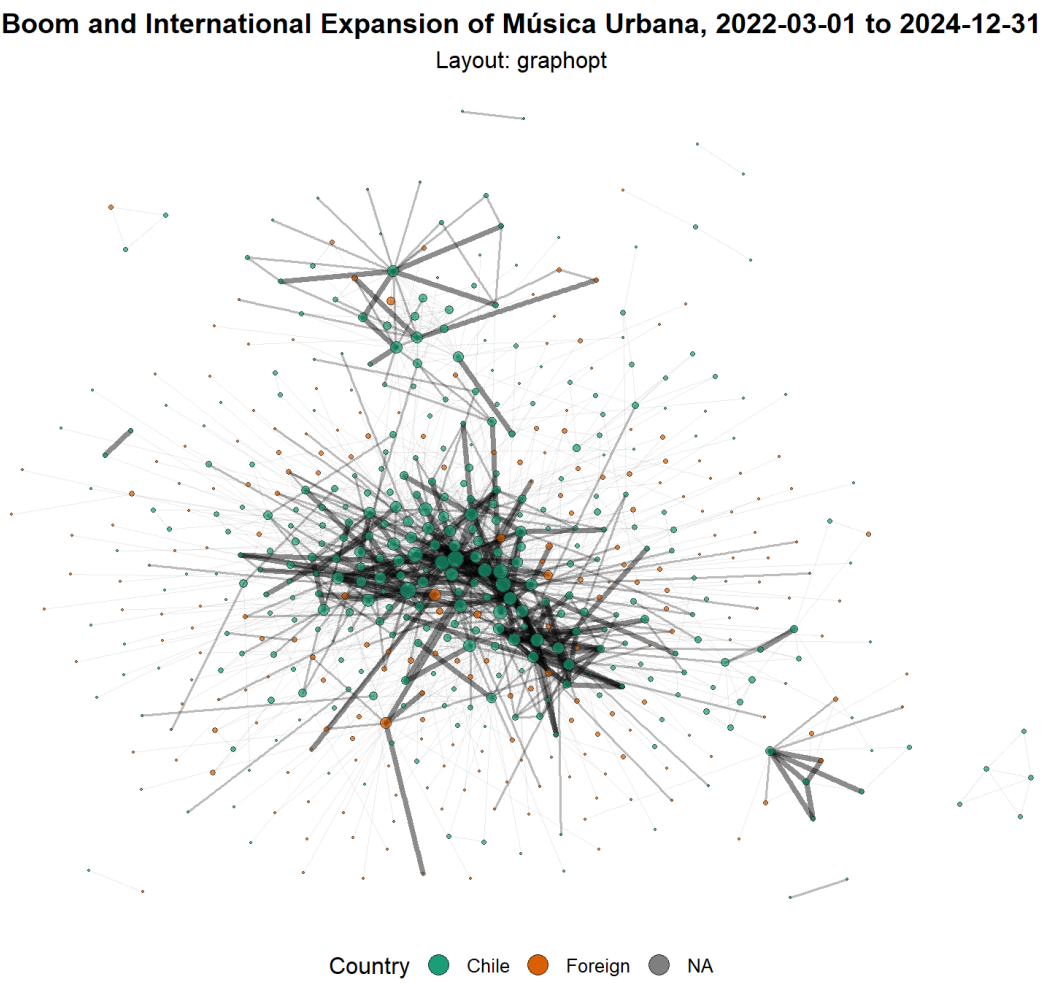


Figure 2. Collaboration network during the boom and international expansion period. Node size is proportional to degree, defined as the number of distinct collaboration partners.

5.1.2 Node Attributes and Network Composition

Figures 3 and 4 describe who participated in the collaboration network in each period. The comparison focuses on country, artist type, gender, label affiliation, and generation.

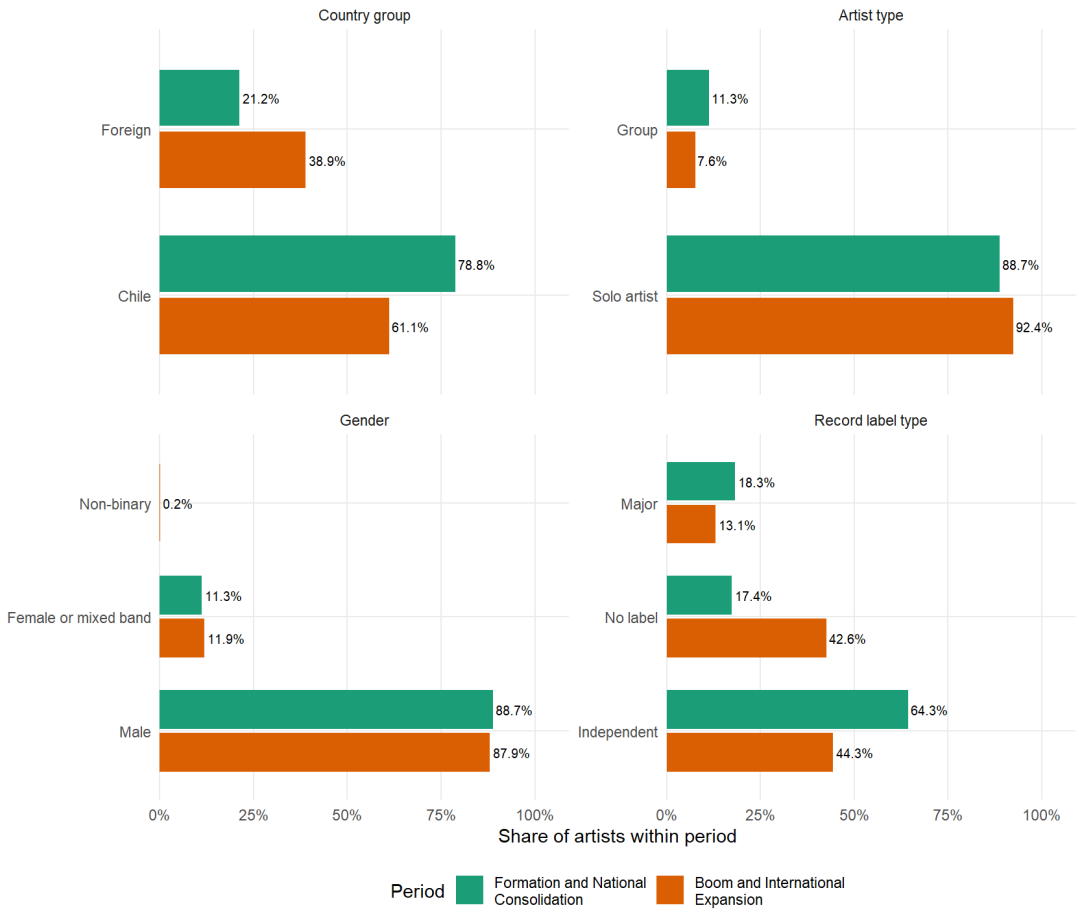


Figure 3. Node composition by period and attribute.

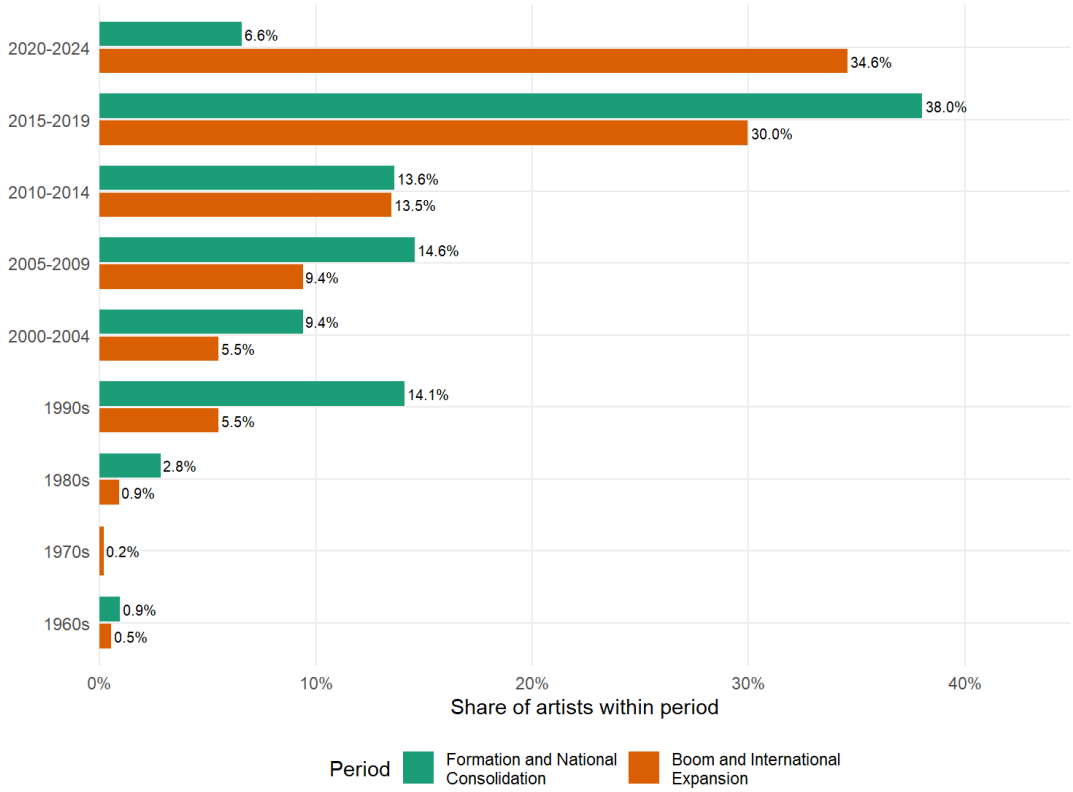


Figure 4. Node generation by period.

The most pronounced compositional change concerns country. Chilean artists remained the majority in both periods, but the foreign share increased from 21.2% to 38.9%. Solo artists also became more prevalent, increasing from 88.7% to 92.4% of nodes, while the relative presence of groups declined.

Gender composition changed little. Male artists accounted for nearly 88% of the network in both periods, while female or mixed-band artists represented approximately 12%. One non-binary artist appeared in the second period. This stability should be read against the strongly masculinised composition of both networks.

Label affiliation and year of inception changed more substantially. Independent artists remained the largest label category, but their share declined from 64.3% to 44.3%, while artists without an identifiable label increased from 17.4% to 42.6%. The proportion associated with major labels decreased slightly.

The distribution by year of inception also differs across periods, with a larger presence of artists beginning between 2020 and 2024 in the second period. This is partly expected given

the dataset's temporal structure, since the later period includes years when newer artists could enter the network. However, it remains analytically relevant because it indicates that the boom and international expansion period includes not only previously established artists, but also a large cohort of recently emerging artists participating in Spotify-visible collaborations.

5.1.3 Weighted Collaboration Patterns

Figure 5 summarises the relative distribution of collaboration intensity across periods. In both periods, most observed collaborations were one-off ties, indicating that occasional collaborations remained the predominant form of artistic interaction. However, the later network also shows a larger proportion of recurrent collaborations. The share of recurrent dyads increased from 17.6% to 25.1%, while recurrent or strong collaborations increased from 5.6% to 8.9%.

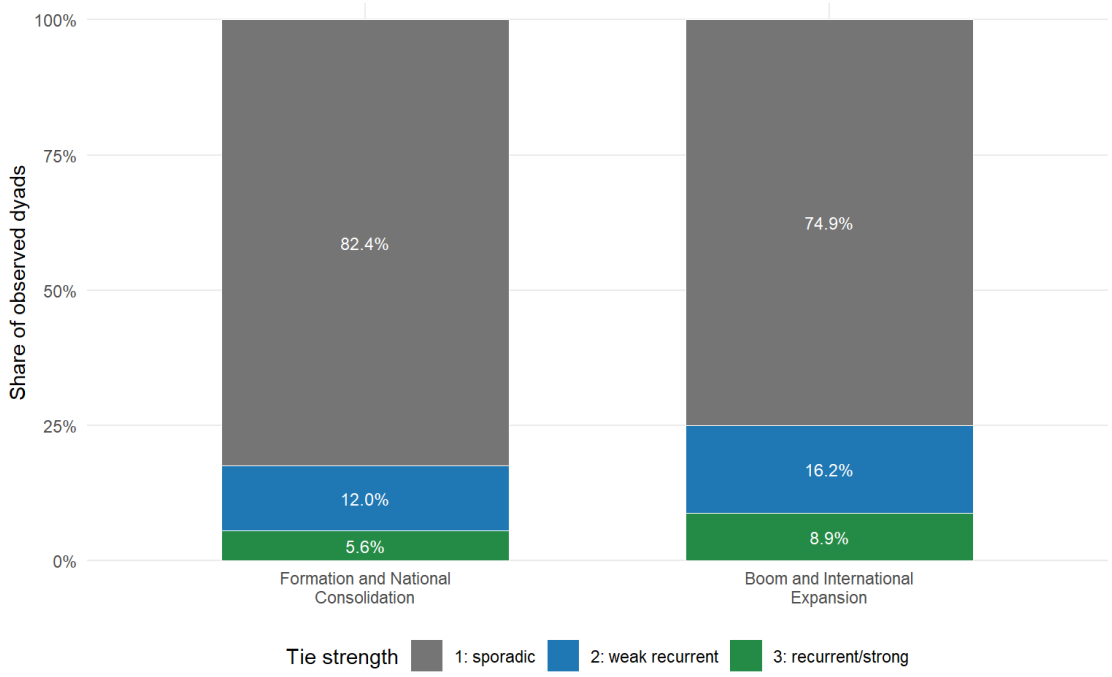


Figure 5. Relative distribution of ordinal collaboration weights by period.

The distribution indicates that network growth was not limited to adding new one-off ties. Sporadic collaborations remained dominant, but repeated collaborations became more common, suggesting greater continuity in a subset of collaborative relationships.

5.1.4 Synthesis of Descriptive Findings

Taken together, the descriptive analyses indicate that the Spotify-visible collaboration network of Chilean *Música Urbana* underwent a substantial structural transformation between the two periods. The network grew rapidly in size, but this growth was accompanied by greater component-level integration rather than fragmentation. Collaboration activity also became more heterogeneous: most artists continued to have few partners, while a smaller set accumulated more ties and stronger collaboration intensity. Closure followed a dual pattern: binary transitivity declined, whereas weighted transitivity increased, indicating stronger cohesion within the closed configurations that remained. The participating population also changed, with a larger foreign share, more varied label affiliations, and a greater presence of artists whose trajectories began in recent years. Overall, the shift from formation and national consolidation to boom and international expansion involved broader participation, stronger integration, and persistent local intensity.

These descriptive patterns also inform the following analytical stages. The combination of lower density, larger connected components, and declining centralisation suggests that expansion did not simply produce a larger version of the first-period network, but a more integrated structure that remained heterogeneous. This motivates the community detection analysis, which examines whether this heterogeneity is organised into meso-level collaboration clusters. At the same time, the stable median degree, rising maximum degree, and increasing collaboration intensity indicate that tie formation is unevenly distributed across artists, while the contrast between declining binary transitivity and increasing weighted transitivity suggests that local cohesion is associated with stronger recurrent ties. These patterns also motivate the ERGM strategy, which evaluates whether generational proximity, Chile-Foreign collaboration, categorical similarity, and low-degree participation are associated with the presence of collaboration ties across the two periods.

5.2 Community Structure

Community detection examines the meso-level organisation of the Spotify-visible collaboration network. Whereas the previous section showed that the network expanded and became more structurally integrated, this section asks how that increasingly connected

collaboration space is internally organised. The analysis bridges descriptive network statistics and the ERGM models by identifying the main collaboration clusters that require interpretation before modelling tie formation.

The analysis is restricted to the largest connected component of each period. This makes community structure comparable across periods and avoids treating disconnected peripheral groups as substantive communities. Leiden is used as the main specification because it is designed to identify well-connected communities and is less prone than Louvain to producing internally disconnected clusters. Louvain is estimated as a robustness check during model selection. For this reason, the substantive interpretation focuses on the selected Leiden partitions.

Formally, the selected Leiden solutions optimise weighted modularity. For a partition c, modularity can be written in simplified form as:

$$Q = (1 / 2m) \sum_{ij} [w_{ij} - \gamma (k_i k_j / 2m)] \delta(c_i, c_j)$$

where w_{ij} is the ordinal collaboration weight between artists i and j, k_i and k_j are their weighted degrees, m is the total edge weight, γ is the resolution parameter, and $\delta(c_i, c_j)$ equals one when two artists are assigned to the same community and zero otherwise. In this dissertation, the procedure identifies groups of artists with more and stronger internal collaborations than would be expected from their weighted connectivity alone.

5.2.1 Selecting the Community Solution

The number of communities was evaluated rather than fixed in advance. For each period and method, we estimated repeated partitions across a common grid of resolution parameters and random seeds. We retained only candidate partitions containing between three and twelve communities. The selection procedure compared modularity, stability across runs, community-size balance, and interpretability. The objective was not to maximise modularity mechanically, but to identify a solution that preserved empirical fit without unnecessary fragmentation.

The first diagnostic uses the best modularity observed for each value of (k). It shows where modularity peaks and whether gains flatten as additional communities are introduced.

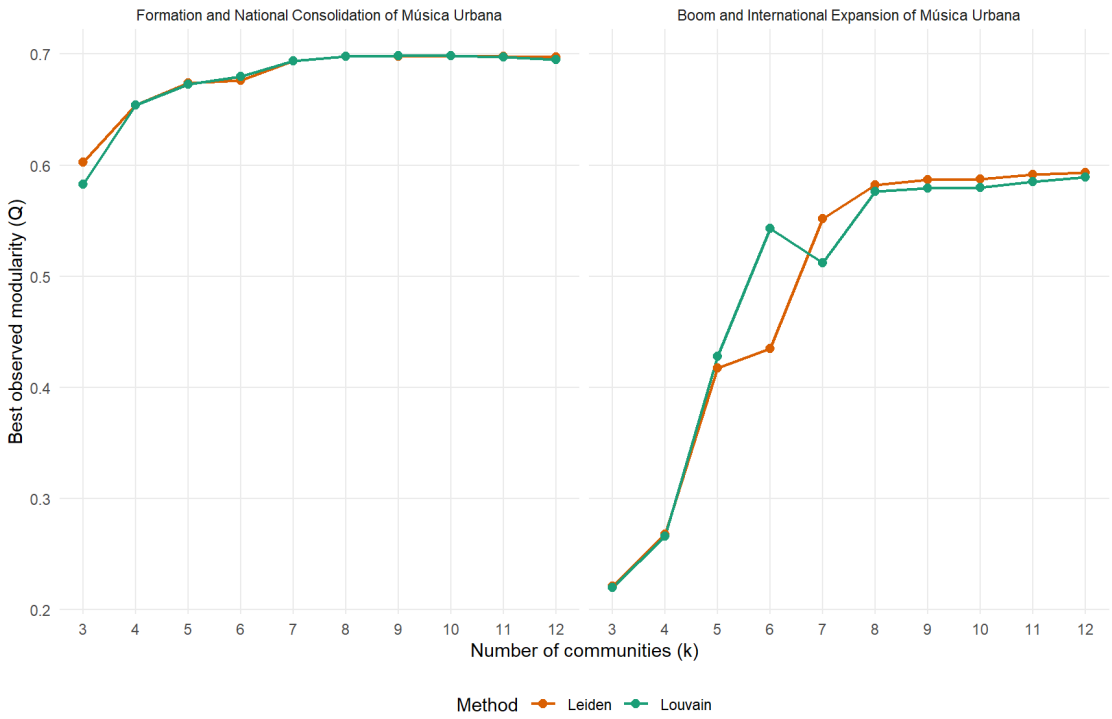


Figure 6A. Modularity curves by period, method, and number of communities.

The second diagnostic uses every retained candidate solution. Narrow boxes indicate stable modularity across resolutions and random seeds, whereas wide boxes indicate that the quality of a given (k) depends more strongly on the particular run.

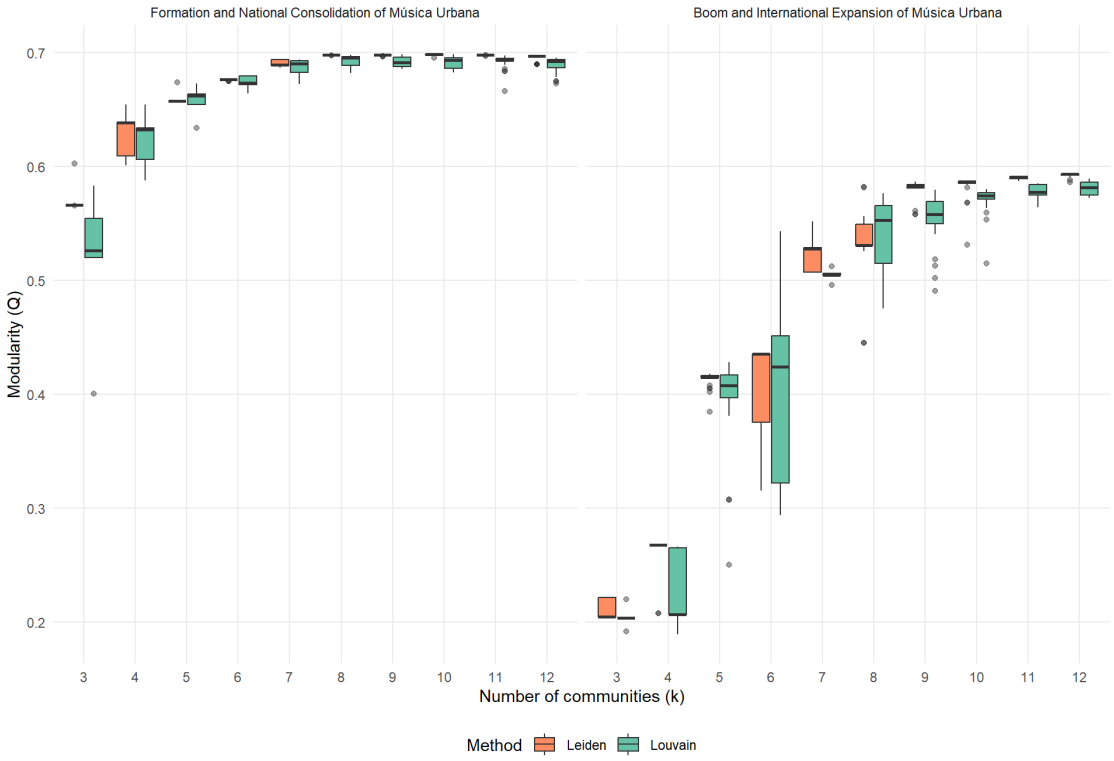


Figure 6B. Distribution of modularity across candidate community solutions.

The full community-size vectors for each candidate value of (k) were exported for audit purposes but are not reported in the main text. The selection is summarised below using the diagnostic curves, boxplots, and final selected solutions.

The final selection combines modularity with parsimony and community-size balance. Seven communities are retained for the Formation and National Consolidation period, while eight communities are retained for the Boom and International Expansion period. In the first period, both methods identify a stable, interpretable partition with seven communities. In the second, modularity rises from seven to eight communities and then enters a region of diminishing returns: for Leiden, the increase from (k=8) to (k=9) is below 0.01, and Louvain shows the same pattern. The agreement across methods indicates that the selected number of communities is not an artefact of a single algorithm. The selected Leiden solutions also avoid very small communities, with the smallest community containing seven artists in the first period and 18 in the second. These values of (k) should be read as analytically useful partitions rather than unique or definitive community structures.

Table 4A. Selected Louvain solution by period.

Period	Selected k	Selected Q	Community sizes	Communities < 5
Formation and national consolidation	7	0.694	47, 37, 34, 26, 21, 19, 7	0
Boom and international expansion	8	0.576	134, 105, 78, 74, 66, 58, 18, 12	0

Table 4B. Selected Leiden solution by period.

Period	Selected k	Selected Q	Community sizes	Communities < 5
Formation and national consolidation	7	0.694	46, 37, 29, 26, 25, 21, 7	0
Boom and international expansion	8	0.582	129, 117, 80, 68, 59, 49, 25, 18	0

All subsequent community descriptions use the selected Leiden solution. Community numbers are ordered by community size within each period and should not be interpreted as equivalent categories across periods.

5.2.2 Structural Characterisation of Selected Leiden Communities

The network visualisations below show the selected Leiden partitions. Node colour identifies community membership, node size represents degree, node shape distinguishes Chilean from foreign artists, and edge width represents ordinal collaboration intensity. The figures should be read as descriptive visualisations of meso-level structure rather than as evidence of the substantive meaning of each community.

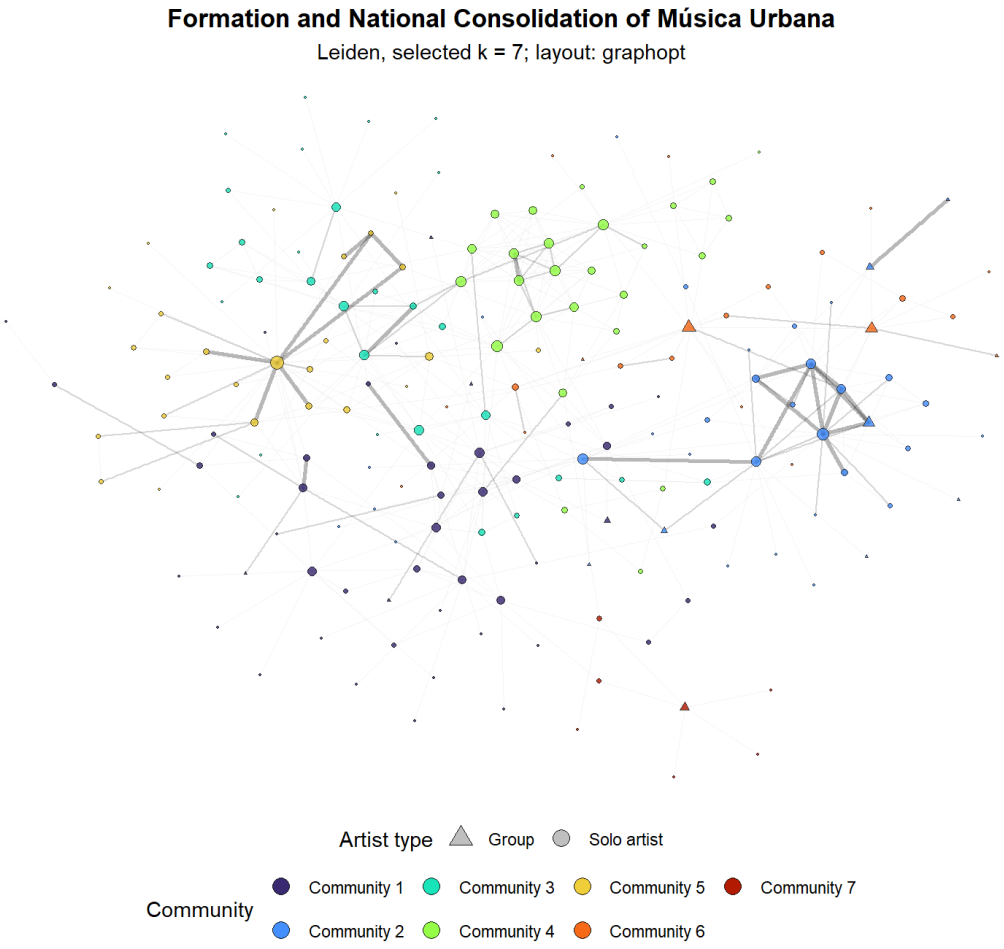


Figure 6A. Leiden communities during the formation and national consolidation period.

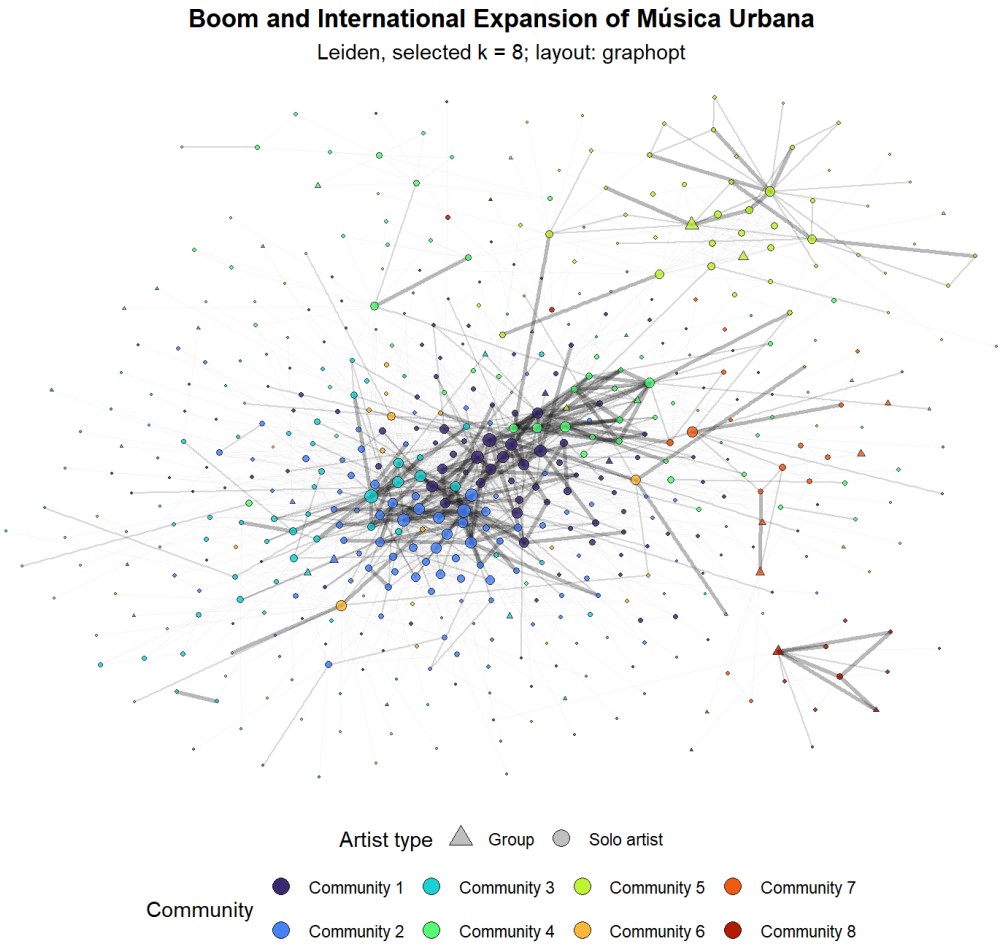


Figure 6B. Leiden communities during the boom and international expansion period.

The visual comparison suggests that network expansion did not eliminate internal differentiation. In the Formation and National Consolidation period, communities appear as smaller, more visually distinct clusters. In the Boom and International Expansion period, communities are embedded within a larger and more integrated connected structure. Thus, the boom period combines two processes: increasing structural integration and persistent meso-level differentiation. The following tables provide a compact compositional reading of these partitions, showing how communities differ in size, collaboration intensity, generational profile, country composition, record-label position, and gender composition.

5.2.3 Community Composition

Tables 5A and 5B characterise the selected Leiden communities separately for each period. The tables combine community size, collaboration intensity, generational profile, foreign-

artist presence, female or mixed-band participation, and the dominant record-label status. Mean ordinal strength is retained as the main indicator of collaboration intensity because it captures repeated collaboration while reducing the influence of extreme dyads.

Table 5A. Selected Leiden communities during Formation and National Consolidation.

Comm.	Nodes	Ord. str.	Dom. gen.	Mean year	Foreign %	Fem./mixed %	Main label
C1	46	3.8	2015-2019	2014	11%	4%	Independent
C2	37	5.7	1990s	2000	14%	27%	Independent
C3	29	4.8	2015-2019	2015	31%	14%	Independent
C4	26	9.2	2015-2019	2015	12%	4%	Independent
C5	25	5.4	2015-2019	2013	33%	0%	Independent
C6	21	3.1	1990s	2002	19%	10%	No label
C7	7	2.0	2015-2019	2015	43%	0%	No label

The first-period solution is especially useful for interpreting the generational and stylistic boundary between rap and *urbano* in Chile when read alongside the historical literature. Communities 2 and 6 can be read together as two closely related clusters of historical Chilean rap. Compared with the other first-period communities, both are marked by older generational profiles: in each case, the dominant cohort corresponds to artists whose trajectories began in the 1990s, whereas most other communities are dominated by artists emerging between 2015 and 2019. This contrast is important because rap and *Música Urbana* do not fully overlap as emic categories in the Chilean context. Although rap is included analytically within the broader delimitation of *Música Urbana* used in this dissertation, many artists, audiences, and scenes distinguish rap–especially boom-bap-oriented rap–from what is more commonly recognised as *urbano*. The community structure reflects this distinction: older rap clusters remain visible within the collaboration network, but they are differentiated from younger clusters associated with trap, reggaeton, mambo, and other contemporary urban sounds.

The comparison between Communities 2 and 6 also reveals internal variation within this historical rap segment. Community 2 displays higher collaboration intensity and a larger presence of female or mixed-band artists. This pattern is partly associated with Ana Tijoux, whose position in the community connects historical rap to collaborations involving women artists, including some foreign artists and artists located outside the core of *Música Urbana*. Community 6, by contrast, is more strongly concentrated around male rap trajectories and

shows no presence of major-label affiliation, whereas Community 2 includes a small but visible major-label component. These differences do not imply a sharp separation between the two clusters, but they help identify variation within the historical rap segment of the first-period network.

The remaining first-period communities are more closely associated with the generational formation of *urbano* as a recognisable category. Community 1 brings together artists linked to early trap and reggaeton trajectories, including Marlon Breeze, Polimá Westcoast, Harry Nach, Tommy Boysen, and Ceaese. Community 4 stands out for its high mean ordinal strength, suggesting a more recurrent collaboration pattern around artists associated with reggaeton and mambo-oriented circuits. Community 5, centred on Pablo Chill-E and Julianno Sosa, combines a younger generational profile with a comparatively larger share of foreign collaborators, pointing to the early articulation of Chilean trap with transnational collaboration circuits. Community 3 and the smaller Community 7 should be interpreted more cautiously, but both reinforce the broader pattern: the first-period network is organised by generational and sonic differences rather than by a single homogeneous urban scene.

Taken together, these patterns suggest that the distinction between rap and *urbano* is not only stylistic but also generational and relational. The emergence of *urbano* in Chile involved consolidating new sounds and collaborative formations that were partly connected to, but also differentiated from, earlier rap trajectories. Community detection therefore provides empirical support for analysing *Música Urbana* as a heterogeneous relational category that gathers multiple musical lineages while preserving internal differences among the artists, sounds, generations, and collaboration circuits that compose it.

Table 5B. Selected Leiden communities during Boom and International Expansion.

Comm.	Nodes	Ord. str.	Dom. gen.	Mean year	Foreign %	Fem./mixed %	Main label
C1	129	8.5	2020-2024	2016	62%	5%	Independent
C2	117	10.2	2020-2024	2018	21%	8%	No label
C3	80	7.4	2020-2024	2015	38%	12%	No label
C4	68	7.6	2015-2019	2013	15%	29%	No label
C5	59	6.7	1990s	2005	27%	14%	Independent
C6	49	3.5	2020-2024	2018	90%	12%	No label
C7	25	5.8	2015-2019	2015	4%	4%	No label
C8	18	4.5	2005-2009	2008	61%	22%	Independent

The second-period solution shows a rearticulation of the collaboration space rather than a simple enlargement of the first-period communities. Community labels are period-specific, and therefore Community 1 in the second period should not be read as the continuation of Community 1 in the first. Instead, a redistribution of artists becomes visible into a larger and more internally differentiated network. The two largest communities are Community 1, with 129 artists, and Community 2, with 117 artists. Community 1 is strongly internationalised, with 62.0% foreign artists and the highest share of major-label affiliation in the period. It includes Chilean figures such as Julianno Sosa, ITHAN NY, El Jordan 23, Pablo Chill-E, Polimá Westcoast, Ak4:20, and Pailita, together with foreign collaborators such as Fuego, JC Reyes, Brray, De La Ghetto, Bad Gyal, KHEA, Dani Flow, El Bogueto, Yung Beef, and Justin Quiles. Community 2, by contrast, is more strongly Chilean and displays the highest mean ordinal strength, suggesting a large local cluster with particularly recurrent collaboration ties around artists such as El BAI, Balbi El Chamako, Marcianeke, Jere Klein, Tunechikidd, Simon la Letra, and Yiordano Ignacio.

The rap-oriented segment remains distinguishable in the second period, especially in Communities 5 and 8. Community 5 brings together several artists from older rap trajectories, including Portavoz, Movimiento Original, Chystemc, Stailok, Ana Tijoux, Semillah Skillz, and Aerstame. Its mean inception year is earlier than that of most other communities, and its dominant generation corresponds to the 1990s. Community 8 is smaller and somewhat later, with a dominant generation of 2005-2009 and artists such as Mamborap, Donexprs, Audio Directo, Doze Ermitaño, and Flor De Rap. This distinction suggests internal differentiation within the rap segment itself: Community 5 is larger and more anchored in historical rap figures, whereas Community 8 appears as a smaller rap cluster with a later generational profile and a comparatively higher share of foreign artists.

The remaining communities are more closely associated with younger *urbano* trajectories. Communities 1, 2, 3, and 6 are dominated by artists whose trajectories began between 2020 and 2024, while Communities 4 and 7 are dominated by the 2015-2019 cohort. Community 4 is notable for the largest presence of female or mixed-band artists in the period, including Vlntna B, KYA, Shirel, AKRIILA, Cami, SOULFIA, and others, and it also includes a visible major-label component. Community 7 is smaller, almost entirely Chilean, and has no major-

label presence, which is consistent with reading it as a more local or alternative *urbano* cluster. Community 6 forms a second internationalised nucleus: it is smaller than Community 1 but has the highest foreign share in the period. It is organised around artists such as Lolo OG, Lauty Gram, Martinwhite, Omar Varela, Salastkbron, DJ Tao, and Marka Akme, and its links to Chilean artists are especially visible through Martinwhite, Sayian Jimmy, Jairo Vera, Gino Mella, Marcianeke, El Joan, Ak4:20, and Standly. This pattern suggests a more specific connection with Latin American urban circuits, especially Argentine and related transnational scenes, rather than the broader international circulation observed in Community 1.

5.2.4 Historical and Theoretical Interpretation

The Leiden results show that community structure is differentiated by size, collaboration intensity, nationality, generation, and organisational position. These patterns do not explain the historical meaning of each community on their own, but they provide a relational map that can be read in dialogue with the historical and ethnographic reconstruction developed earlier in the dissertation and in previous work on Chilean *Música Urbana* (Muñoz-Tapia et al., 2025; Muñoz-Tapia and Pinochet-Cobos, 2025). In this sense, community detection helps assess whether the category's historical formation leaves observable traces in Spotify-visible collaboration networks.

The communities should therefore not be interpreted as bounded genres. They combine artists with different profiles, suggesting that collaboration clusters are formed through the articulation of stylistic, generational, industrial, and transnational relations. This is consistent with relational and assemblage approaches to genre, which understand musical categories as historically situated configurations of sounds, actors, institutions, media, and classificatory practices rather than fixed stylistic containers (Brackett, 2016; Drott, 2013). It also connects with White's and Crossley's arguments that categories and networks are mutually implicated: categories orient recognition and affiliation, while relations help stabilize categories as socially effective formations (Crossley, 2025a; White, 2008).

The comparison between periods clarifies this point. In the Formation and National Consolidation period, the network is organised around a visible generational and stylistic

boundary between historical rap and younger *urbano*-oriented formations. This pattern resonates with previous research showing that the emergence of *Música Urbana* in Chile involved a contested process of categorisation structured by the split between rap and trap, the development of reggaeton, and the production of generational subjectivities (Muñoz-Tapia and Pinochet-Cobos, 2025). The first-period communities make this boundary observable at the relational level: rap-oriented clusters are associated with older cohorts and boom-bap trajectories, while trap, reggaeton, and mambo-oriented clusters are dominated by younger artists.

In the Boom and International Expansion period, this distinction does not disappear, but it is reorganised within a larger and more heterogeneous collaboration network. Rap remains identifiable through specific communities, while larger *urbano*-oriented clusters are dominated by artists emerging between 2015 and 2024. At the same time, internationalisation appears as a selective process rather than a uniform opening of the network. One nucleus is broader and more industry-visible, combining Chilean artists with collaborators from multiple international scenes; another is smaller and more specifically connected to Latin American *urbano* circuits. This suggests that the expansion of *Música Urbana* involved both the persistence of internal generational boundaries and the formation of regional collaborative horizons.

Taken together, the two periods show that *Música Urbana* operates as a glocal hypergenre in a relational sense (Buch, 2013, 2018; Muñoz-Tapia et al., 2025). It gathers heterogeneous musical lineages without erasing their internal differences. Historical rap, trap, reggaeton, mambo, and other *urbano* sounds become part of the same analytical network, yet they remain differentiated by generation, sonic orientation, collaboration intensity, country composition, and organisational position. Community detection therefore extends previous historical and ethnographic research by showing how categorical tensions are also enacted through collaboration among artists on songs. The next section turns from identifying clusters to modelling whether selected endogenous processes and artist attributes help account for the formation of collaboration ties across the two periods.

5.3 ERGM

5.3.1 Binary ERGM Strategy

The final modelling stage uses binary Exponential Random Graph Models (ERGMs) to examine which mechanisms are associated with the presence of collaboration ties in the Spotify-visible networks. The dependent network is defined as:

0 = no observed collaboration
1 = at least one observed collaboration

In this analysis, the dependent variable is therefore a binary artist-by-artist collaboration network estimated separately for each period. A tie takes the value 1 when two artists share at least one credited track collaboration within the period, and 0 otherwise. This specification asks a conservative question: given the set of artists retained in each period, which endogenous and attribute-based mechanisms are associated with whether two artists collaborate at least once?

Weighted collaboration remains central to the descriptive and community analyses, where tie intensity is analysed directly. For the ERGM stage, however, the binary specification was retained because it provided a more stable and comparable strategy across the two periods than the valued alternatives explored during modelling. The models are estimated separately for the Formation and National Consolidation period and the Boom and International Expansion period, following the standard ERGM logic of modelling the observed network as one realisation from a distribution of possible networks (Koskinen and Snijders, 2012; Krivitsky, 2012).

The binary ERGM estimates the probability of observing a collaboration network Y as a function of selected structural and attribute-based statistics:

$$P(Y = y) = [1 / \kappa(\theta)] \exp \{ \theta^T g(y) \}$$

where y is the observed collaboration network, g(y) is a vector of network statistics, θ is a vector of model parameters, and κ(θ) is the normalising constant. In this dissertation, the model assesses whether collaboration ties are associated with generational proximity, Chile–foreign ties, categorical similarity, and endogenous network structure.

5.3.2 Model Specification

The model sequence is progressive and intentionally parsimonious. M1 establishes a baseline for tie formation after accounting for the network's structural design constraints. M2 adds the two mechanisms most directly connected to the dissertation argument: generational proximity and Chile-Foreign collaboration. M3 incorporates categorical homophily by gender and artist type. M4 adds degree(1), an endogenous term that accounts for the large number of artists with only one observed collaboration partner.

Two structural offsets are included throughout the sequence and are not interpreted substantively. First, degree(0) is offset because the final analytic networks contain no isolates. Second, foreign-foreign dyads are offset because foreign or non-core artists enter the data only through ties to the Chilean *Música Urbana* core. The Chile-Foreign term therefore compares Chile-Foreign ties with Chile-Chile ties within the eligible dyad space, rather than modelling all possible international collaborations.

Table 6. Binary ERGM sequence evaluated for the dissertation.

Model	Specification	Purpose
M1: Baseline	edges + offsets	Baseline model for tie formation after accounting for structural design constraints.
M2: Generation + transnational tie	M1 + year difference + Chile-Foreign tie	Adds generational proximity and the distinction between Chile-Chile and Chile-Foreign ties.
M3: Attribute homophily	M2 + same gender + same artist type	Adds categorical similarity among artists.
M4: Low-degree participation	M3 + degree(1)	Selected model; adds endogenous structure associated with low-degree participation.

5.3.3 Progressive Model Comparison

Tables 7A and 7B compare the same four models in each period. Coefficients are shown with standard errors below them; AIC, BIC, and log likelihood are reported at the bottom of each table. The sequence is not designed to maximise fit separately for each period, but to evaluate whether successive blocks of theoretically motivated terms improve a comparable specification across the two networks. The purpose is therefore not to identify the most

complex possible model, but to retain a stable specification that can be interpreted alongside the descriptive and community results.

Table 7A. Binary ERGM results for formation and national consolidation.

Term	M1: Baseline	M2: Generation + transnational tie	M3: Attribute homophily	M4: Low-degree participation
Edges	-3.96***	-2.98***	-4.16***	-3.20***
	(0.05)	(0.07)	(0.24)	(0.20)
Degree(1)				2.96***
				(0.24)
Year difference		-0.11***	-0.11***	-0.09***
		(0.01)	(0.01)	(0.01)
Chile-Foreign tie		-1.00***	-1.02***	-0.46***
		(0.16)	(0.16)	(0.13)
Same gender			0.73***	0.38**
			(0.18)	(0.14)
Same artist type			0.61***	0.32*
			(0.16)	(0.14)
AIC	4033.08	3824.88	3799.84	3648.90
BIC	4041.10	3848.96	3839.96	3697.05
Log Likelihood	-2015.54	-1909.44	-1894.92	-1818.45

Table 7B. Binary ERGM results for boom and international expansion.

Term	M1: Baseline	M2: Generation + transnational tie	M3: Attribute homophily	M4: Low-degree participation
Edges	-4.39***	-3.34***	-4.75***	-4.30***
	(0.03)	(0.04)	(0.13)	(0.11)
Degree(1)				2.84***
				(0.19)
Year difference		-0.09***	-0.08***	-0.07***
		(0.01)	(0.01)	(0.00)
Chile-Foreign tie		-1.64***	-1.68***	-1.00***
		(0.07)	(0.07)	(0.07)
Same gender			0.98***	0.76***
			(0.09)	(0.08)
Same artist type			0.62***	0.42***

Term	M1: Baseline	M2: Generation + transnational tie	M3: Attribute homophily	M4: Low-degree participation
			(0.10)	(0.08)
AIC	17005.20	15991.75	15808.74	15556.52
BIC	17015.18	16021.67	15858.61	15616.37
Log Likelihood	-8501.60	-7992.87	-7899.37	-7772.26

The comparison supports M4 as the principal specification. Adding generational proximity and Chile-Foreign ties improves fit relative to the baseline; adding categorical homophily further improves fit; and adding degree(1) improves AIC, BIC, and log likelihood in both periods. The signs of the main effects are consistent across periods: year difference and Chile-Foreign ties are negative, while same gender, same artist type, and degree(1) are positive. The more negative Chile-Foreign coefficient in the boom period suggests that the increased presence of foreign artists did not make cross-national collaboration generalised across the eligible dyad space. For this reason, M4 is retained as the main comparative ERGM, while M1-M3 are interpreted as progressive comparison models.

5.3.4 Selected Model Diagnostics

M4 is selected because it is estimated successfully in both periods, improves fit over M3, and provides the same interpretable specification for the two networks. This comparability is important because the objective is to evaluate whether similar mechanisms organise collaboration across periods, rather than to maximise fit separately for each network. GOF and MCMC diagnostics are therefore reported only for M4.

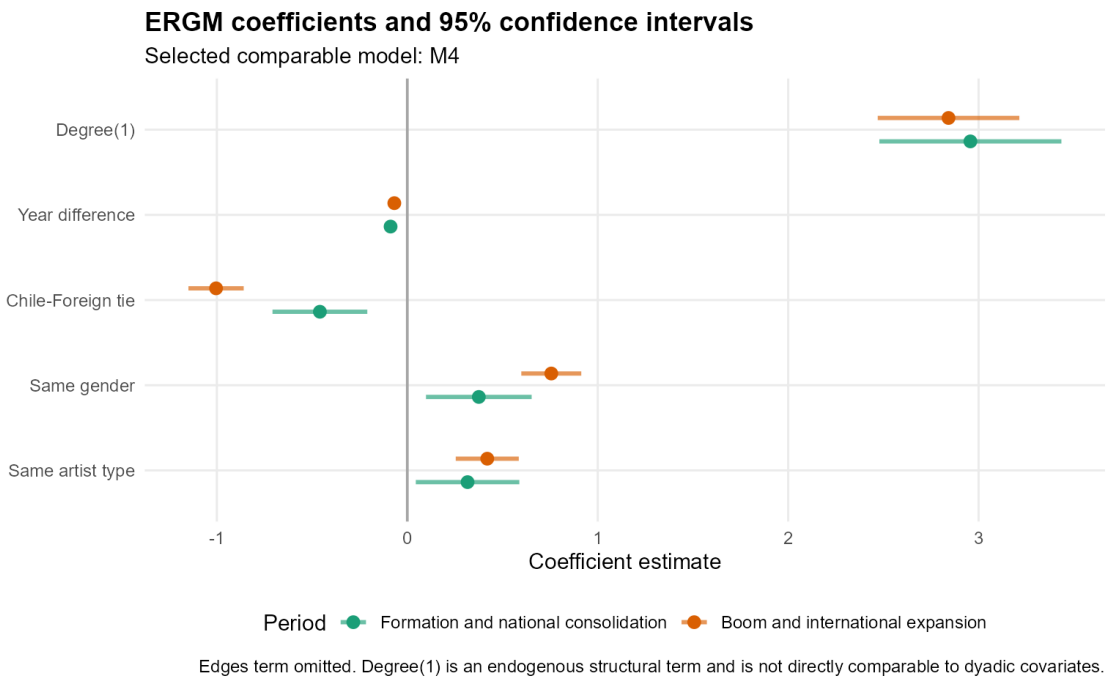


Figure 8. Coefficient estimates and 95% confidence intervals for M4 by period. The edges term is omitted for readability.

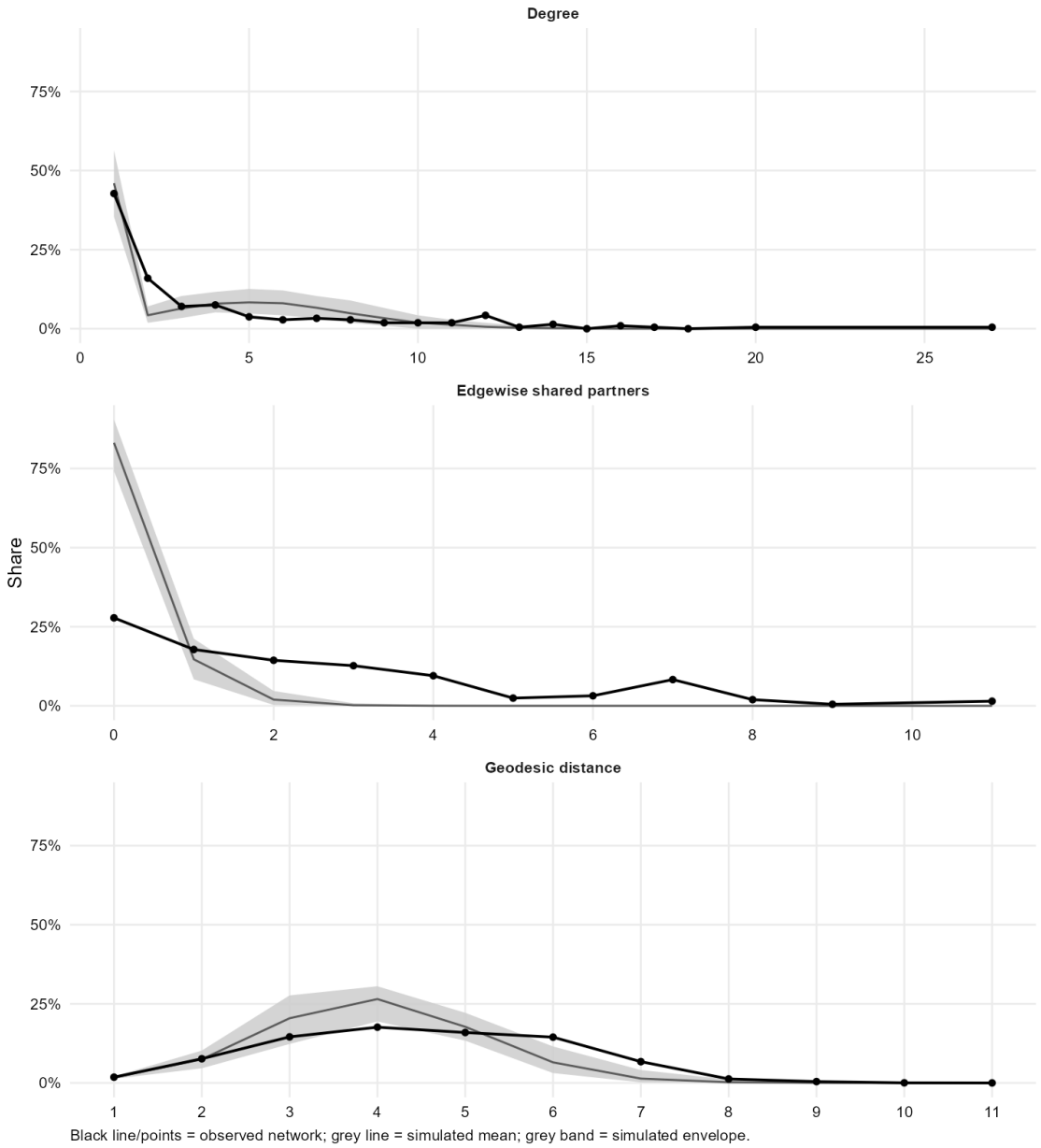


Figure 9A. GOF diagnostics for M4 in the formation and national consolidation period. Black line/points represent the observed network; grey line and band represent simulated networks.

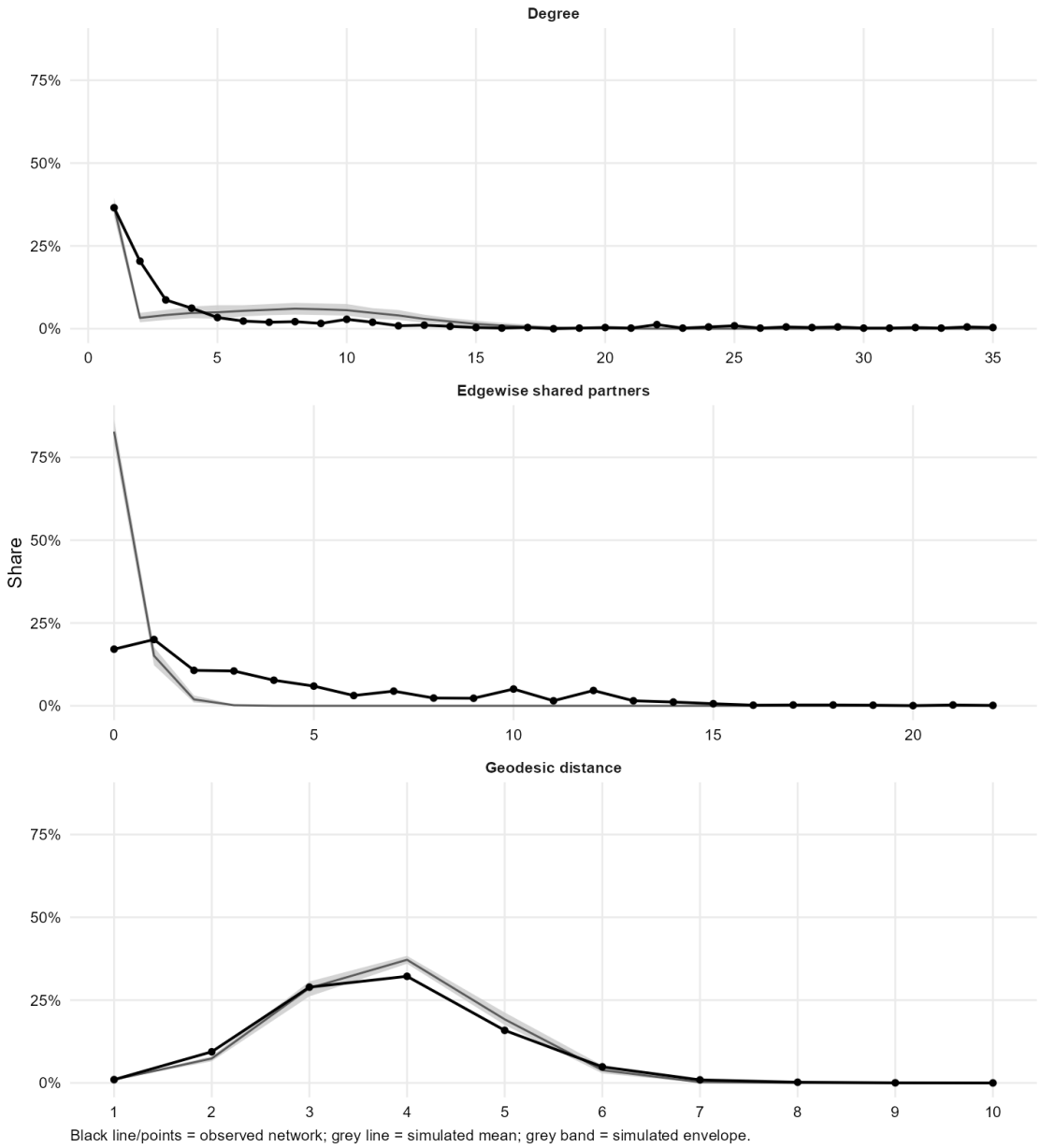


Figure 9B. GOF diagnostics for M4 in the boom and international expansion period. Black line/points represent the observed network; grey line and band represent simulated networks.

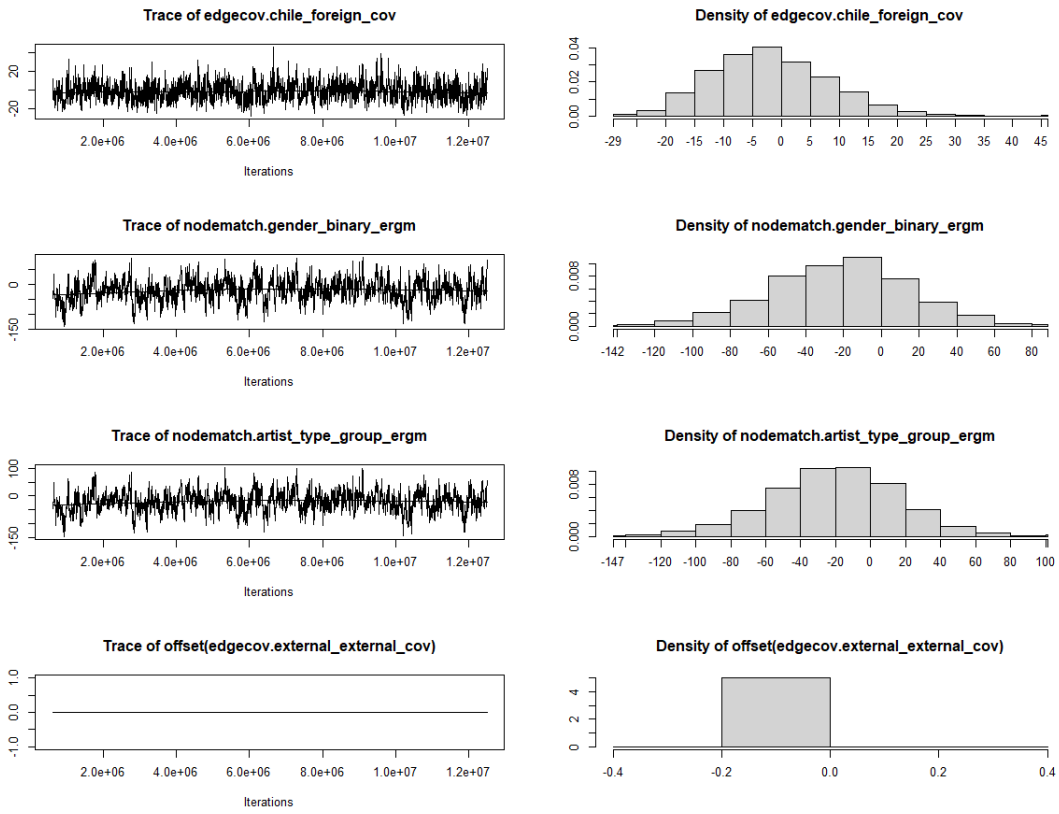


Figure 10A. MCMC diagnostics for M4 in the formation and national consolidation period.

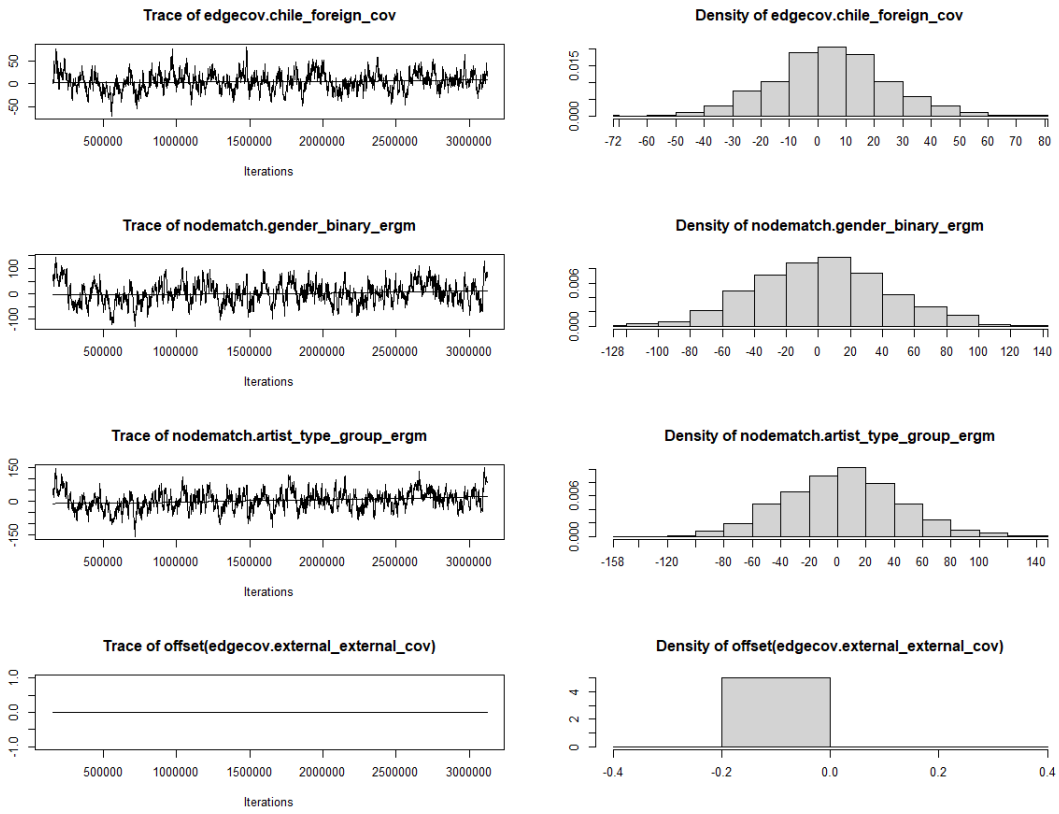


Figure 10B. MCMC diagnostics for M4 in the boom and international expansion period.

The GOF diagnostics suggest that M4 provides a reasonable approximation of broad connectivity patterns in both networks, especially geodesic distances and the degree distribution after incorporating degree(1). The main limitation concerns edgewise shared partners: in both periods, the observed networks contain more dyads with multiple shared partners than the simulated networks. M4 therefore underestimates shared-neighbourhood structure. The MCMC diagnostics do not show obvious explosive drift, supporting the use of M4 as a cautious comparative model, but the GOF results indicate that local closure should not be interpreted through this model alone. This is why closure is interpreted jointly through weighted transitivity, community detection, and the ERGM sensitivity checks.

5.3.5 Closure Sensitivity Checks

Because of this limitation, we estimated additional GWDSP specifications as sensitivity checks. These models are not treated as part of the main progressive sequence. In the formation period, one GWDSP specification slightly improved AIC but did not improve BIC

relative to M4. In the boom period, analogous specifications were computationally unstable and did not provide a comparable alternative.

GWDSP is therefore retained only as a sensitivity check. This decision is methodological rather than substantive: it should not be interpreted as evidence that closure is absent from the collaboration network. Local cohesion is interpreted primarily through weighted transitivity and community detection, while the ERGM is used to evaluate more stable mechanisms of tie formation.

5.3.6 Substantive Interpretation

Because the models are estimated on separate networks, coefficient comparisons across periods are interpreted cautiously, focusing on direction, consistency, and broad relative change. The negative edges coefficient in both periods reflects network sparsity: most eligible dyads do not collaborate. Its more negative value during the boom should not be read as a decline in collaboration, but as a consequence of the expansion of the dyadic opportunity structure.

Generational proximity is consistently associated with collaboration. Artists whose trajectories began closer in time are more likely to collaborate, and this association remains after adding categorical homophily and low-degree structure. The coefficient is slightly less negative in the boom period, consistent with a larger network in which more cohorts coexist, but the effect remains important. This supports the historical interpretation of Chilean *Música Urbana* as a field marked by generational differentiation.

Chile-Foreign ties are negative in both periods, with a stronger negative coefficient during the boom. This does not contradict the descriptive and community results showing increased internationalisation. Rather, it indicates that cross-national ties remain less likely than Chile-Chile ties within the eligible dyad space. Internationalisation therefore appears as a selective expansion concentrated in particular sectors of the network, rather than as a generalised weakening of national structuring. Community detection identifies where internationalised collaboration is concentrated, while the ERGM shows that this pattern does not amount to generalised cross-national mixing across all eligible dyads.

degree(1) is positive and strong in both periods, indicating that artists with only one observed collaboration partner are not residual cases but part of the network structure. This is consistent with the descriptive results: the boom expands the network substantially, but many artists still enter the Spotify-visible collaboration structure through a single observed partner. Expansion therefore reproduces peripheral modes of participation at a larger scale.

Same-gender collaboration is positive in both periods and stronger in the boom model, but this result should be interpreted in relation to the strongly masculinised composition of the network. Since male solo artists constitute the large majority of nodes, same-gender collaboration partly reflects the field's demographic structure rather than gender operating as the principal axis of community formation. At the same time, gender is not irrelevant: the community analysis suggests that the visibility of women and mixed-gender collaboration depends heavily on specific figures and localised configurations, such as the community around Ana Tijoux.

Same-artist-type collaboration is also positive in both periods but remains a secondary effect. The result is consistent with a collaboration field centred mainly on individual artists, while groups and collectives occupy more specific positions. Finally, the remaining misfit for edgewise shared partners suggests that local cohesive formations should not be reduced to a single ERGM closure mechanism. The ERGM therefore complements the descriptive and community analyses by identifying robust associations with tie formation while preserving a cautious interpretation of local clustering.

6. Discussion and Conclusion

This dissertation has examined Chilean *Música Urbana* not simply as a genre label, but as a historically situated category whose social effectiveness can be observed through patterns of collaboration. The findings support a relational interpretation of musical categorisation: categories do not merely describe pre-existing artistic worlds; they also orient recognition, affiliation, and interaction. In this sense, the analysis extends the argument that categories and networks are mutually constitutive (Crossley, 2025a; White, 2008). *Música Urbana*

appears not only as a classificatory device, but as a relational formation through which artists, scenes, and trajectories become connected.

A central contribution of the study is methodological and interpretive. The network analysis does not replace historical interpretation; rather, it triangulates and extends it. Muñoz-Tapia and Pinochet-Cobos (2025) show that the emergence of *Música Urbana* in Chile involved transformations in rap, digital self-production, platform visibility, and generational understandings of what counted as *urbano*. That historical account is crucial because it gives specific meaning to the generational divide observed in the network. The distinction between older rap trajectories and newer *urbano* clusters is not simply an age difference or a statistical partition; it corresponds to a locally situated process through which artists, audiences, and scenes came to distinguish rap, trap, reggaeton, and other urban sounds. The present analysis extends that account by showing that this categorical and generational divide is also organised through collaboration ties. In this sense, the relationship between categories and networks is retroactive: categories orient recognition and affiliation, while repeated collaborations help reinforce and stabilise those categories. The main contribution, then, lies in showing how Chilean *Música Urbana* became relationally organised between formation and national consolidation and boom and international expansion: it expanded, became more structurally integrated, and developed through selective internationalisation.

In that sense, this dissertation articulates two analytical emphases that often appear disconnected: musical genre and art worlds. Genre foregrounds classification, categorisation, recognition, and symbolic struggle, while art worlds highlight the cooperative chains and relational infrastructures that make artistic production possible. The Chilean case shows that these dimensions are not separate: the genre label becomes meaningful through the relations that enact it, while those relations are oriented by shared categories (Crossley, 2025a; White, 2008). Methodologically, a more quantitative Social Network Analysis can be combined with qualitative strategies such as interviews and participant observation because each captures a different dimension of the same cultural process. Historical and ethnographic work clarifies the category's meanings, while network analysis shows how those meanings become relationally stabilised through collaboration (Muñoz-Tapia and Pinochet-Cobos, 2025).

This approach is important for cultural network analysis because Spotify collaboration data are not treated as a transparent representation of a scene. Instead, they serve as an empirical trace of a broader process of categorisation, circulation, and artistic coordination. The analysis combines platform data with historical and interpretive knowledge of the Chilean case, in line with approaches that understand musical categories as historically situated assemblages rather than fixed stylistic containers (Brackett, 2016; Drott, 2013). Social Network Analysis therefore makes visible one dimension of the relations through which categories acquire social force, while accounting for interpretation of the historical and cultural context.

One of the clearest empirical findings concerns internationalisation. Compared with the formation and national consolidation period, the boom and international expansion period includes a larger share of foreign artists, and the community analysis identifies clusters with stronger international presence. At first glance, this could suggest a general transnationalisation of Chilean *Música Urbana*. The ERGM results complicate that interpretation. Chile-Foreign ties remain less likely than Chile-Chile ties, and this negative association is stronger during the second period. Internationalisation should therefore not be understood as a homogeneous opening of the entire collaboration network. Rather, it appears as a selective process concentrated in particular relational sectors. Foreign artists become more visible and more numerous, but cross-national collaboration does not become a generalised pattern across the eligible dyad space.

This result speaks to broader debates on streaming and global music circulation. Hesmondhalgh (2025) argues that streaming has produced more multipolar patterns of musical visibility while preserving unequal structures of platform ownership, infrastructural control, and market power. Chilean *Música Urbana* fits this ambivalent landscape. The boom period allowed local artists to connect with broader Latin American circuits, but this expansion remained selective and uneven. The findings therefore nuance celebratory accounts of platform globalisation: streaming may increase transnational visibility, but it does not automatically produce evenly distributed transnational collaboration.

Network expansion also did not produce symmetric integration. Between the two periods, the network shows a sharp increase in nodes, edges, and total collaborations, and the largest

component contains a greater share of the network. In this sense, the collaboration network became more structurally integrated. Yet this integration coexists with persistent inequality in participation. In terms of collaboration opportunities, expansion increased the number of possible positions in the network without distributing access to collaboration evenly across artists. Most artists continue to have few collaborators, and the ERGM shows a strong positive degree(1) effect in both periods. The second period therefore did not eliminate peripheral participation; it reproduced it at a larger scale.

This pattern is especially relevant in the context of the featuring economy and platform visibility. Collaboration can function as a mechanism of circulation, exposure, and legitimacy, but a single collaboration does not necessarily imply stable integration into a collaborative world. The network grows through both one-off ties and recurrent partnerships, producing a structure that is simultaneously larger, more integrated, and highly heterogeneous. The transition between periods should therefore be understood not only as growth, but also as uneven relational incorporation.

Gender appears as a secondary but important dimension of the analysis. The descriptive results show that the network is strongly masculinised in both periods. The ERGM identifies a positive same-gender effect, particularly during the boom, but this coefficient must be interpreted cautiously because male solo artists constitute the overwhelming majority of nodes. Gender homophily partly reflects the field's demographic structure. The community analysis further suggests that gender does not organise the main divisions of the network, which are more clearly differentiated by generation, musical trajectory, country composition, and international orientation. However, gender is not irrelevant. The visibility of women and mixed-gender collaboration appears to depend heavily on specific figures and localised configurations. The community around Ana Tijoux is especially important in this respect, since it connects historical rap trajectories with women artists, foreign artists, and artists located partly outside the core of *Música Urbana*. Gender therefore operates less as a global organising principle than as an uneven and localised dimension of collaboration.

The study also has methodological implications for analysing cultural categories through networks. Descriptive measures show expansion and integration, community detection identifies meso-level clusters, and the ERGM evaluates mechanisms associated with tie

formation. No single method captures the full process. This is especially clear for local cohesion: weighted transitivity and community detection indicate cohesive local formations, while the ERGM does not fully reproduce edgewise shared partners. Collaboration clusters therefore require both formal modelling and historical-cultural interpretation.

The approach also clarifies the limits of Spotify-visible data. The dataset captures collaborations that are visible on the platform, not the entire set of musical, social, and organisational relations that constitute the scene. It therefore offers a partial but analytically useful trace of collaboration. By combining these data with historical knowledge and manually validated artist attributes, the dissertation avoids treating platform data as self-sufficient evidence.

Taken together, the findings show that the formation, consolidation, and boom of Chilean *Música Urbana* cannot be understood only as a story of stylistic change, platform visibility, or international success. It is also a process of relational transformation. The category gathers heterogeneous musical trajectories, but it does not erase their internal differences. Generational boundaries, selective internationalisation, uneven participation, and localised gender visibility all remain visible within the collaboration network. The dissertation therefore contributes to a cultural-relational understanding of musical categories: categories acquire social force not only through discourse, markets, or platforms, but through the patterned relations that connect artists and make a musical world observable across time.

6.1 Methodological Limitations

The networks analysed in this dissertation represent collaborations that are visible in Spotify metadata, not the complete universe of musical relations. They capture credited co-participation on released tracks but exclude informal creative exchanges, uncredited production and songwriting, live performance, friendship, management, and other relations that may shape collaboration. The network boundary is therefore substantive and platform-specific rather than a census of all artists or interactions in Chilean music.

The sharp increase between periods should also be interpreted as a combination of scene expansion and changing platform visibility. More artists may have entered the field, but digital distribution, access to Spotify profiles, metadata coverage, and formal crediting

practices also expanded. The later network consequently documents the growth of the Spotify-visible collaboration field rather than providing a direct estimate of growth in the underlying population of musicians.

Artist classification combines Spotify genre metadata with manual validation. This improves the situated validity of the *Música Urbana* category, but it also introduces interpretive decisions that cannot be reduced to a platform label. The relational boundary mitigates this issue by requiring external artists to connect directly to the selected Chilean *Música Urbana* network, and the audit confirms that every final edge satisfies this rule.

7. Abbreviations

- API: Application Programming Interface
- DSP: Digital Service Provider
- ERGM: Exponential Random Graph Model
- GOF: Goodness of Fit
- MCMC: Markov Chain Monte Carlo
- SNA: Social Network Analysis

8. Statements and Declarations

8.1 Ethical Considerations

The network analysis uses publicly available and platform-derived metadata about artists and released tracks. It does not involve interviews, experiments, private communications, or direct interaction with human participants as part of the dissertation's network analysis. The analysis reports aggregate patterns and uses artist names only where they are already public and analytically necessary for interpreting collaboration networks.

8.2 Declaration of Conflicting Interests

The author declares no potential conflicts of interest with respect to the research, authorship, or publication of this dissertation.

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